Safety Factor

Ride Engineer · Corbin Park

15 DAYS · 61 STOPS · GRADE 11 · MIDWAY

THE OPENING CARD

Corbin Park lost its certificate in October and the season opens in three weeks. Seven rides have to get it back — a coaster, a wheel, a drop tower, a pirate ship, a carousel, a bumper floor and a log flume — and the signature on all seven documents is yours. You are the ride engineer, hired in February. What you have inherited is eleven notebooks in one hand: forty-one years of settings, timings and turns of a nut from a mechanic who retired last autumn, with not one line of working anywhere in them. Delia Marsh, who owns the park, points out that eleven million people have ridden on those numbers unhurt. Marcus Vey, who inspects for the county, calls that a record of having got away with it. A hundred and six days of season pay for the other two hundred and fifty-nine.

11 notebooks and no working

BEFORE THE DAY — GREET

The park has been shut for two years and the crew is new

Corbin Park closed in October two years ago and the team reopening it has never worked together. Vey, the county ride inspector, wants you known to as many of them as you can before the first inspection, because a ride is signed off by one person on the word of another, and this fortnight is nothing but that.

PLAN CARD 130W

Monday, and the season opens in three weeks. Corbin Park lost its certificate in October and seven rides have to get it back before the gates open, which is your signature on seven documents. Delia Marsh, who owns the park, has laid out Alf Brennan's 11 notebooks on the workshop bench: 41 years of settings, timings and turns of a nut, in 1 hand, with not 1 line of working anywhere in them. Marcus Vey, the county inspector, has read the same notebooks and calls them a record of having got away with it. Today you take the first three numbers the park runs on and find out where they come from. 106 days of season pay for the other 259, and nothing opens on a setting nobody can derive.

BRIEFING 19W

Every setting in this park is in one man's handwriting and none of them has a calculation behind it.

WHAT YOU DECIDE 11W

Find out what the park actually knows about its own rides.

1.1 What 42 metres of fall is worth

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · DERIVE

SCENE 33W

The tower drops its carriage 42 metres before the brakes begin. An old engineering notebook gives the bottom speed as 'about 29' and nothing else. The safety engineer wants the working written underneath.

GUIDE 69W

You build the derivation a line at a time. The starting relation is on the rail, and at each step you choose the expression the line above actually gives you. Every wrong branch is algebraically legal, so you cannot find the answer by spotting the malformed one — you have to know which operation the previous line licenses. Work down to the speed at the bottom of the fall.

ASK 20W

Get the speed at the end of 42 metre free fall, from the constant-acceleration relations rather than from the notebook.

BEHIND THE BUTTON 162W

Why derive it rather than look it up. The notebook says 'about 29' and nothing else, which cannot be checked, cannot be corrected for a different drop height, and cannot be defended to an inspector. A derivation is the same number plus the reason it is that number, and the reason is what transfers to the next ride.

What the starting relation says. $v^2 = v_0^2 + 2a\Delta x$ is a statement about constant acceleration with the time taken out of it. That is exactly the convenient form here, because the question asks about a distance fallen and says nothing about how long the fall lasted.

What is being assumed. Released from rest makes v_0 zero, and treating a as g assumes the fins do nothing until they engage and that drag on the carriage is small over 42 metres. Both are stated on the card rather than hidden, because the number is only as defensible as the assumptions it was computed under.

TAKES AS READ

an object in free fall accelerates at about 9.8 metres per second every second · the constant-acceleration relations connect speed, acceleration and distance · acceleration as the rate a velocity changes, whichever part of it changes — taken as read · newton's second law as the definition of a net force — taken as read

VERDICT 90W

The old notebook's 'about 29' was right, but the engineer signing the safety certificate needs a defensible chain. The relation $v^2 = v_0^2 + 2a\Delta x$ fits because the distance is known and time is not. Starting from rest leaves $v = \sqrt{(2g\Delta x)}$, with no mass in the result. That is why objects in free fall share the same acceleration. The square matters: doubling the fall distance raises speed only by $\sqrt{2}$, about 41%. The same relation will run backward tomorrow when the brake removes that speed over a known distance.

TAKEAWAY 17W

Choose the constant-acceleration relation that contains the quantities you know and omits the one you do not.

1.2 Reading a circuit as energy

THE COASTER · COASTER STATION · A PERSON · PROTOCOL

SCENE 36W

Marta Kovač walks the track every morning. On the workshop wall she has four places chalked out. They are the lift hill, top of the first hill, bottom of the first drop, and final braking section.

GUIDE 65W

Four places on the circuit, and four statements about energy. Pair them by asking two things of each. Where is the energy sitting, in height or in motion? And is any being added or removed there? Only one place on the whole circuit adds energy. One removes it and never gives it back. That is why no later hill reaches the height of the first.

ASK 15W

Match each point on the circuit to what is true of the train's energy there.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

lifting something stores energy that can be given back · a moving body carries energy that depends on the square of its speed

VERDICT 84W

The lift motor adds mechanical energy while the train climbs. After the top of the first hill, height and speed trade energy back and forth. Friction drains some into heat all around the circuit, so later hills cannot reach the height of the first hill. The foot of the first drop is fast because much of the stored height has become kinetic energy. The final braking section removes the remaining mechanical energy deliberately. Reading the circuit this way separates energy transfers from energy losses.

TAKEAWAY 17W

Mechanical energy trades between height and motion, while motors add it and friction or brakes remove it.

1.3 60 people and the same 5 seconds

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A RIDE · CHOICE

SCENE 42W

Sam Idowu has run the pirate ship for 20 years. During low-angle maintenance swings, a full boat and an empty one both take 5.9 seconds. He wants to know why loading changes the force on the pivot bearings but not that timing.

GUIDE 68W

All four options explain a period that does not move when the boat fills. Ask of each whether it names a cancellation or an accident. The restoring pull is the boat's own weight, and the inertia resisting it grows in the same proportion. So they cancel, and length and gravity are what remain. Note the limit too: this holds at small angles, and the ride swings 48 degrees.

ASK 10W

Why does adding riders leave the low-angle period essentially unchanged?

BEHIND THE BUTTON 195W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a pendulum has a restoring force toward its lowest point · Newton's second law connects net force, mass and acceleration

VERDICT 76W

The restoring force comes from the boat's own weight. Add mass and that force grows in the same proportion as the inertia resisting acceleration. Those factors cancel, so the small-angle period does not depend on mass. Length and local gravity remain. This statement has a limit: the simple period formula assumes a small swing angle. At the ride's 48-degree operating swing, the period is a little longer. Loading still barely changes it, but amplitude now matters.

TAKEAWAY 21W

> A variable can disappear from a motion when it scales both the restoring force and the inertia in the same way.

END OF THE DAY 14W

A number that works is not the same as a number somebody can justify.

DAY 2 OF 15

What the brake has to take out

BEFORE THE DAY — TRIAL

Walk the park before you certify anything on it

Seven rides, one midway, and the layout is the syllabus: tower, coaster, carousel, wheel, bumper floor, ship, flume. Walk the whole park once. Every one of them will be argued about this fortnight, and an inspection report written by somebody who has not stood under the ride reads exactly like one.

PLAN CARD 141W

Tuesday, and the tower is first on Vey's list because it is the ride with the least between a rider and the ground. Yesterday's derivation says the carriage reaches the fins at 29 metres a second; the fins have 9 metres to take that back. Rafael Santos, who maintains the brakes, points out that nothing in the system is adjustable — the magnets make more force the faster the fins move — and Vey's reply is that a brake nobody can adjust is still a brake nobody has measured. Today you turn a stopping distance into an acceleration, get a pressure out of a depth at the flume, and settle what a moment arm is under the wheel. 4.7 times gravity is a number a fit adult takes in their seat and a number you would not put a child through.

BRIEFING 13W

The tower's brake has nine metres to remove everything the fall put in.

WHAT YOU DECIDE 10W

Turn a stopping distance into a force on a person.

2.1 9 metres to take back 29

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · BALLPARK

SCENE 33W

The carriage reaches the brakes at 28.7 metres a second and stops 9 metres later. The tower technician wants that turned into an acceleration before anyone argues about whether the stop feels acceptable.

GUIDE 59W

Five numbers, and two of them belong to the fall rather than the stop: the free-fall height above, and gravity. Ask of each whether the braking depends on it. And note the square on the entry speed. Halve the stopping distance and the average deceleration doubles, which is why a small rise in entry speed costs so much brake.

ASK 8W

Estimate the average deceleration in the braked section.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the same constant-acceleration relation runs backwards for slowing down · an acceleration can be quoted as a multiple of g

VERDICT 70W

The stopping relation is the same constant-acceleration equation used on the drop, but solved for a instead of v. With 28.7 m/s removed over 9.0 m, the average deceleration is about 45.8 m/s^2, or 4.7 g. The square on speed is the design lesson. Halving the stopping distance doubles the average deceleration. Likewise, a modest increase in entry speed demands much more braking force and leaves less room for error.

TAKEAWAY 18W

A stopping distance and a speed are enough to fix the acceleration, with no time measured at all.

2.2 What nine metres of water pushes with

THE LOG FLUME · FLUME PUMPHOUSE · A RIDE · BALLPARK

SCENE 33W

The water surface in the elevated tank is 9.4 metres above its outlet. Kelechi Abara has a pressure gauge on that outlet and an old notebook value that does not match the reading.

GUIDE 65W

Five numbers, and two of them describe the tank rather than the water above the outlet: the diameter, and the flow through it. Ask of each whether pressure depends on it. A narrow pipe of the same height gives the same pressure, which is why the width is offered and not needed. The outlet answers to the depth above it, however big the tank looks.

ASK 10W

Estimate the pressure above atmospheric pressure at the tank outlet.

BEHIND THE BUTTON 106W

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Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

pressure in a liquid grows with depth below the surface · water has a density of about 1000 kilograms per cubic metre · newton's second law as the definition of a net force — taken as read

VERDICT 77W

Hydrostatic pressure comes from the weight of liquid above each unit area. It depends on density, gravity and depth, not on tank width. For 9.4 m of water, the pressure above atmospheric pressure is about 92 kPa. A narrow vertical pipe of the same height would produce the same pressure. That is why tank diameter is a distractor here. The outlet hardware responds to the pressure at its depth, even when the tank around it looks enormous.

TAKEAWAY 15W

Depth sets pressure, and the width of the tank has nothing to do with it.

2.3 The number people get wrong

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · CHOICE

SCENE 38W

Jan Novák has a six-metre crane arm raised 30 degrees above horizontal near the Ferris wheel. A load hangs vertically from its end. His lifting plan and the shop sketch disagree about the distance used to calculate torque.

GUIDE 63W

Four distances, and only one of them is perpendicular to the line the load pulls along. Ask of each which direction it runs in. The load hangs straight down, so the arm that matters runs horizontally. The steel is six metres long and not all of it counts. Raising the arm shortens the horizontal distance, so the same load twists the pivot less.

ASK 20W

The crane arm is 6 metres long and angled up at 30°. Which distance sets the torque about the pivot?

BEHIND THE BUTTON 195W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

a force applied further from a pivot has a greater turning effect · a force can be drawn as a line extending in both directions

VERDICT 74W

Torque uses the perpendicular distance from the pivot to the line along which the force acts. The load hangs vertically, so the needed arm is horizontal. For a 6 m crane arm at 30 degrees, that arm is 6 cos30°, or about 5.2 m. The full steel length is not the lever arm. Raising the arm shortens the horizontal arm, so the same hanging load produces less torque about the pivot. That distinction matters.

TAKEAWAY 22W

> A turning effect is fixed by where the line of the force passes, not by the shape of the thing carrying it.

2.4 One g, all the way down

THE DROP TOWER · DROP TOWER CONTROL · A PERSON · HOLD

SCENE 42W

The tower's magnetic brake is set from the console and the car has to come down inside the deceleration band the certificate names. The car's mass changes with every load, and the brake does not know how many people are in it.

GUIDE 46W

Hold the deceleration inside the band on the console. The band narrows toward the bottom, because the last stretch is where a rider's neck takes whatever the brake does. The brake current is your control, and every change in load changes what the same current produces.

ASK 10W

Hold the deceleration inside the band all the way down.

BEHIND THE BUTTON 137W

Why the same brake gives a different ride. A net force divided by mass is the acceleration, so the same braking force decelerates a heavy car less than a light one. The brake current sets the force; the riders set the mass; the certificate is written about the acceleration, which is the ratio.

Why nothing here is a step. A magnetic brake's force depends on speed, so as the car slows the force falls and the deceleration falls with it. Holding a deceleration means raising the current continuously through the stop rather than setting it once.

Why the band tightens near the bottom. The car is slowest there, the brake is weakest there, and the rider's tolerance is unchanged. It is the one part of the drop where the margin has to be made rather than inherited.

TAKES AS READ

a force applied to a mass produces an acceleration · acceleration as the rate a velocity changes, whichever part of it changes — taken as read

VERDICT 139W

The certificate limits the deceleration. The console sets the braking force. Between them sits the mass in the car, and that changes on every cycle. Twelve adults and four children is a different vehicle from four adults. A current that gave a comfortable stop this morning gives a harsher one when the car is light. That is the second law as an operating rule rather than an equation. There is a second difficulty. A magnetic brake's force depends on how fast the car moves through the field, so it fades as the car slows. Hold the current steady and the deceleration decays away with it. The current has to be raised through the stop. And the band narrows at the bottom, where the brake is weakest and the rider's neck is no stronger than it was at the top.

TAKEAWAY 17W

The same force is a different ride, because what the rider feels is force divided by mass.

END OF THE DAY 16W

Every force in this park is a mass times an acceleration somebody has to work out.

Where the energy went

BEFORE THE DAY — FOLLOW

The coaster walk-down, at the pace it is actually done

The track is walked from the lift hill to the brake run, and the person doing it stops at every joint and support. Stay with them. The stops are the inspection; the walking between them is just walking, and somebody hurrying it will have seen the track without having looked at it.

PLAN CARD 132W

Wednesday, and Kovač has three seasons of station-return speeds on a clipboard: 6.1 metres a second in her first year, 5.2 this spring, measured the same way each time with the same wheel. Nothing about the hill has changed, which means the difference is friction and the difference is energy the ride no longer has. Marsh reads it as wear and a bearing job. Vey reads it as a circuit whose margins are all computed from a drawing made in 1974. Today you close the energy books on one lap, work out what a circular ride does to the direction of a velocity, and count momentum through a collision on the bumper floor. Whatever the loop needs, it has to come out of what is left at the top of the hill.

BRIEFING 17W

The train comes back to the station slower every year and nobody has measured by how much.

WHAT YOU DECIDE 13W

Find the losses on the circuit, and what they leave for the loop.

3.1 The lap that does not come back

THE COASTER · COASTER STATION · A RIDE · DERIVE

SCENE 41W

The train leaves the top of the first hill at 26 metres and reaches the bottom of the first drop 2 metres above the station. Kovač's wheel says 21.4 metres a second at the bottom, and an ideal circuit says more.

GUIDE 65W

You build the derivation a line at a time, choosing at each step the expression the line above gives you. Every wrong branch is legal algebra, so the malformed-line trick will not help. Work the energy statement down to the speed at the foot of the drop with no losses — and then say what Kovač's measured 21.4 metres a second implies about the difference.

ASK 23W

Work the energy statement down to the speed at the foot of the drop, and say what the measured speed then tells you.

BEHIND THE BUTTON 178W

What the starting statement says. Energy at the crest equals energy at the bottom plus whatever friction took. Written out, that is height and speed at the top against height and speed at the bottom, with one term for the losses. Nothing about the shape of the track appears, which is why energy is the tool of choice here rather than forces.

Why the mass cancels, and when it does not. Every term in the ideal statement carries m, so it divides out and the ideal speed is independent of how loaded the train is. The friction term is not proportional to m in the same way, so the moment losses enter, the loaded and empty trains stop agreeing.

What the gap is worth. The difference between the ideal speed and 21.4 is the energy the circuit lost on the way down, expressed as a speed. Turning that into joules and then into a percentage is how a maintenance decision gets made: a lap that does not come back is a wheel, a bearing or a brake dragging.

TAKES AS READ

stored energy is weight times height, and kinetic energy is half the mass times the speed squared · friction does negative work along the whole path

VERDICT 80W

Start with mechanical energy at the top of the first hill and at the foot of the drop. Mass appears in every term and divides out. That is why loading the train does not change the ideal speed from the same height. The ideal result is about 21.8 m/s, while the measured value is 21.4. The difference is energy lost to friction. That loss can be tracked around the circuit, especially near the loop where the speed margin is smallest.

TAKEAWAY 16W

> The gap between the ideal speed and the measured one is a measurement of the friction.

3.2 Going round at a steady speed

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A PERSON · CHOICE

SCENE 33W

The carousel turns at eight revolutions a minute and has done so since the previous engineer set it. Bisi Adeyemi is wiring a new controller and wants the outer-horse acceleration put on paper.

GUIDE 63W

Four options, and two of them agree on the size and disagree on the direction. Ask what is changing when a speed stays constant. Velocity carries a direction as well as a size, and on a circle the direction changes every instant. So decide which way the acceleration points before worrying about its value. One option needs a mass the question never gives.

ASK 16W

An outer horse goes round at a constant eight revolutions a minute. What is its acceleration?

BEHIND THE BUTTON 195W

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TAKES AS READ

velocity has a direction as well as a size · acceleration is the rate at which velocity changes · velocity against speed: direction is part of the quantity — taken as read

VERDICT 66W

Speed can stay constant while velocity changes because velocity also contains direction. On a circle, that direction changes every instant. The resulting acceleration points toward the centre and has size v^2/r. Here 5.2 m/s on a 6.2 m radius gives about 4.4 m/s^2. The platform and pole supply the inward force. A rider feels the seat pushing inward while their inertia makes the experience seem outward.

TAKEAWAY 15W

> Acceleration describes any change in velocity, including a change in direction when speed stays constant.

3.3 100 collisions an hour

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · CHOICE

SCENE 36W

Wei Chen has accelerometers in three bumper cars and 400 impacts logged from last season. In one clean test, a moving car hits a stationary car and the pair travel together for an instant after contact.

GUIDE 58W

Four options, and each keeps or discards two quantities. Ask of each what the collision does to the outside world in a tenth of a second. Almost no outside push arrives, which protects one of the two. The cars leave together, deformed and warm, which does not protect the other. That difference is what soft bumpers are for.

ASK 21W

A 320 kg car at 3.1 m/s strikes a stationary 290 kg car and they move off together. What is conserved?

BEHIND THE BUTTON 195W

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TAKES AS READ

momentum is mass times velocity, counted with direction · kinetic energy depends on the square of the speed · velocity against speed: direction is part of the quantity — taken as read · newton's second law as the definition of a net force — taken as read

VERDICT 68W

Total momentum stays the same because the pair receives almost no external impulse during the brief collision. Kinetic energy does not. The cars leave together at about 1.63 m/s, carrying roughly 810 J of kinetic energy after starting with about 1,540 J. The missing mechanical energy became deformation, sound and heat. That distinction separates an inelastic collision from an elastic one and explains why soft bumpers reduce rebound.

TAKEAWAY 15W

A collision can preserve one bookkeeping quantity while transforming another into heat, sound and deformation.

3.4 What 42 metres of fall is worth

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · CALLBACK · DERIVE

SCENE 33W

The tower drops its carriage 42 metres before the brakes begin. An old engineering notebook gives the bottom speed as 'about 29' and nothing else. The safety engineer wants the working written underneath.

GUIDE 69W

You build the derivation a line at a time. The starting relation is on the rail, and at each step you choose the expression the line above actually gives you. Every wrong branch is algebraically legal, so you cannot find the answer by spotting the malformed one — you have to know which operation the previous line licenses. Work down to the speed at the bottom of the fall.

ASK 20W

Get the speed at the end of 42 metre free fall, from the constant-acceleration relations rather than from the notebook.

BEHIND THE BUTTON 162W

Why derive it rather than look it up. The notebook says 'about 29' and nothing else, which cannot be checked, cannot be corrected for a different drop height, and cannot be defended to an inspector. A derivation is the same number plus the reason it is that number, and the reason is what transfers to the next ride.

What the starting relation says. $v^2 = v_0^2 + 2a\Delta x$ is a statement about constant acceleration with the time taken out of it. That is exactly the convenient form here, because the question asks about a distance fallen and says nothing about how long the fall lasted.

What is being assumed. Released from rest makes v_0 zero, and treating a as g assumes the fins do nothing until they engage and that drag on the carriage is small over 42 metres. Both are stated on the card rather than hidden, because the number is only as defensible as the assumptions it was computed under.

TAKES AS READ

an object in free fall accelerates at about 9.8 metres per second every second · the constant-acceleration relations connect speed, acceleration and distance · acceleration as the rate a velocity changes, whichever part of it changes — taken as read · newton's second law as the definition of a net force — taken as read

VERDICT 90W

The old notebook's 'about 29' was right, but the engineer signing the safety certificate needs a defensible chain. The relation $v^2 = v_0^2 + 2a\Delta x$ fits because the distance is known and time is not. Starting from rest leaves $v = \sqrt{(2g\Delta x)}$, with no mass in the result. That is why objects in free fall share the same acceleration. The square matters: doubling the fall distance raises speed only by $\sqrt{2}$, about 41%. The same relation will run backward tomorrow when the brake removes that speed over a known distance.

> **TAKEAWAY** 17W
>
> Choose the constant-acceleration relation that contains the quantities you know and omits the one you do not.

END OF THE DAY 17W

Friction takes a fixed amount of energy each lap, and it takes it out of the margin.

Eight turns a minute

PLAN CARD 131W

Thursday, and Alf Brennan walked up from his house this morning to be asked why the swing carousel runs at 10 turns a minute and not 11. His answer is the one it always is: at 11 the chains came out too far and it felt wrong. Vey has been polite about this for three days. Today you derive the angle the chains hang at from the speed and the radius, put a number on the energy in the ship's swing, and balance the wheel on paper. Marsh is right this afternoon, and it is worth saying out loud: the derived angle lands within a degree of the chalk mark Brennan made in 1991, and 2 days of an engineer's time have confirmed what a mechanic knew by looking at it.

BRIEFING 16W

Brennan set the swings by how the chains hung, and nobody has ever computed the angle.

WHAT YOU DECIDE 9W

Check a forty-one-year-old setting against the physics behind it.

4.1 Why the chains come out

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · DERIVE

SCENE 35W

The swing carousel turns at ten revolutions a minute. Its chains are 4.5 metres long and hang from a hub 3.0 metres from the axis. An old chalk mark shows where the seats once reached.

GUIDE 58W

Two force equations start you off — one for the horizontal pull and one for the vertical. Build down a line at a time to a relation for the angle the chains hang at. Then read the relation itself: the useful part of this derivation is which quantities survive to the end and which cancel on the way.

ASK 19W

Work the two force equations down to a relation for the angle, and say what the angle depends on.

BEHIND THE BUTTON 166W

Why two equations. A seat on a chain travels a horizontal circle at constant speed, so the net force on it points at the axis and has no vertical part. Splitting the tension into its horizontal and vertical components gives one equation for each direction, and dividing one by the other eliminates the tension nobody measured.

What falls out, and why it matters at the gate. The mass cancels: a heavy rider and a light rider hang at the same angle, which is why the operator does not sort the queue by weight and why the old chalk mark is a mark about speed rather than about who was sitting there.

What the angle does depend on. The rotation rate and the geometry — chain length and hub radius, which together set the radius of the circle the seat travels in. So a carousel running slightly fast shows it in the angle, and the chalk mark is a record of what fast used to look like.

TAKES AS READ

a body moving in a circle needs a net force toward the centre · the tension in a chain acts along the chain

VERDICT 78W

The chain tension has two jobs. Its vertical component balances the rider's weight, while its horizontal component supplies the inward force for circular motion. Dividing those force equations removes both the unknown tension and the rider's mass. What remains links the angle to speed and the radius of the seat's circular path. Because the seat swings outward, that radius grows with the angle and must be solved with it. At ten rpm, the result is about 31 degrees.

TAKEAWAY 16W

Resolving one force into components can remove unknowns and reveal which variables actually control the motion.

4.2 What the swing is worth at the bottom

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A PERSON · BALLPARK

SCENE 33W

The pirate ship reaches 48 degrees on a full cycle. Idowu wants the speed through the bottom before the bearing work is quoted. The loaded boat's centre sits 8.6 metres below the pivot.

GUIDE 61W

Five numbers, and two of them belong to other questions: the pivot-to-centre length, and the period. Ask of each whether the speed at the bottom depends on it. Energy answers to the change in height and not to the shape of the path. So the height risen is the term that matters, and the length is how it was worked out.

ASK 12W

Estimate the speed of the boat through the bottom of its swing.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

stored energy is weight times the height risen · the height risen by a pendulum comes from the geometry of the swing

VERDICT 72W

The ship is a body falling along a curved path. Mechanical energy depends on the change in height, not on the shape of that path. At 48 degrees, the centre of mass rises about 2.85 m above the bottom. Turning that height into kinetic energy gives about 7.5 m/s. The tower reaches the same square-root form by kinematics. The shared result is useful because both situations convert gravitational potential energy into motion.

TAKEAWAY 20W

A swing is a fall along a curve, and the height it falls through is the only part that matters.

4.3 Loading one side of a wheel

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · BALLPARK

SCENE 37W

Passengers board passenger cabins at the bottom, six at a time. During loading, one side of the wheel can carry much more passenger mass than the other. Priya Raman wants the worst temporary imbalance put on paper.

GUIDE 57W

Five numbers, and one of them is the radius of the wheel rather than the average distance the loaded cabins sit from the axle. Ask which distance the torque uses. An empty wheel nearly balances, because opposite cabins cancel. What the drive and brake have to hold is not the heaviest load but the most uneven one.

ASK 15W

Estimate the out-of-balance torque about the axle with four passenger cabins loaded on one side.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

torque is a force times the perpendicular distance from the axle · a torque can be balanced by another torque acting the other way

VERDICT 74W

An empty wheel is nearly balanced because opposite passenger cabins cancel one another's torques. Loading breaks that symmetry before the wheel is full. Four passenger loads on one side produce about 177 kN·m using their average perpendicular distance from the axle. The worst loading case is therefore not the heaviest total load. It is the most uneven arrangement. That temporary imbalance is what the drive, brake and hub must resist while passengers are boarding.

TAKEAWAY 15W

A wheel is balanced by its own symmetry, and loading is what breaks the symmetry.

4.4 Speed, or the turn

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · BELT

SCENE 40W

The certificate needs every complaint on the log classified before it is written. Some of what the riders describe is the ride going fast. The rest is the ride turning, and a rider cannot tell the difference from the seat.

GUIDE 50W

Two bins. What a rider feels in a turn is the seat pushing them toward the centre, and it depends on speed squared over radius — so a tight slow turn can beat a fast straight. Sort each complaint on whether it is about how fast or about how sharply.

ASK 12W

Send each complaint to the bin that says what actually produced it.

BEHIND THE BUTTON 95W

Why riders cannot tell. A body has no sense that measures speed. What it senses is acceleration, which is what a seat, a rail or a harness has to supply. On a straight at constant speed there is nothing to feel at all, however fast it is.

Why the radius does the work. The acceleration in a turn is the speed squared divided by the radius, so halving the radius doubles it while nothing about the speed has changed. That is why the tightest bend, not the fastest section, is where a ride hurts people.

TAKES AS READ

a body feels the force pushing it, not its speed · velocity against speed: direction is part of the quantity — taken as read

VERDICT 169W

A rider has no organ that measures speed. What the body senses is acceleration, which is what the seat, the harness or the rail has to supply — so a straight run at constant speed produces no sensation at all however fast it is, and a tight bend at walking pace can be unpleasant. The acceleration in a turn is the speed squared over the radius, which is why the radius carries so much of the blame: halve it and the rider feels twice as much with the ride going exactly as fast as before. That is the classification the certificate needs. Complaints about being thrown against the door, crushed into the seat or unable to lift an arm are all the turn, and they are fixed by geometry or by a speed limit through that geometry. Complaints about the ride being too fast are usually about something else — noise, duration, or the drop — and slowing the whole ride to answer them fixes nothing anybody actually felt.

TAKEAWAY 15W

What a rider feels is acceleration, and a turn supplies it without going any faster.

4.5 Reading a circuit as energy

THE COASTER · COASTER STATION · A RIDE · CALLBACK · PROTOCOL

SCENE 36W

Marta Kovač walks the track every morning. On the workshop wall she has four places chalked out. They are the lift hill, top of the first hill, bottom of the first drop, and final braking section.

GUIDE 65W

Four places on the circuit, and four statements about energy. Pair them by asking two things of each. Where is the energy sitting, in height or in motion? And is any being added or removed there? Only one place on the whole circuit adds energy. One removes it and never gives it back. That is why no later hill reaches the height of the first.

ASK 15W

Match each point on the circuit to what is true of the train's energy there.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

lifting something stores energy that can be given back · a moving body carries energy that depends on the square of its speed

VERDICT 84W

The lift motor adds mechanical energy while the train climbs. After the top of the first hill, height and speed trade energy back and forth. Friction drains some into heat all around the circuit, so later hills cannot reach the height of the first hill. The foot of the first drop is fast because much of the stored height has become kinetic energy. The final braking section removes the remaining mechanical energy deliberately. Reading the circuit this way separates energy transfers from energy losses.

> **TAKEAWAY** 17W
>
> Mechanical energy trades between height and motion, while motors add it and friction or brakes remove it.

END OF THE DAY 18W

Confirming a good number is not wasted work; it is the difference between a setting and a limit.

DAY 5 OF 15

The seat, the pole and the ram

PLAN CARD 118W

Friday, and the first week ends with three parts nobody has a number for. Adeyemi will not wire the new controller until somebody tells her what force an outer horse's pole carries at eight turns a minute. Santos has a hydraulic gate on the flume with a two-centimetre ram driving a nine-centimetre one and a note in Brennan's hand reading "plenty". Vey has spent the morning reading yesterday's derivation and has signed nothing. Today you get the sideways force on a carousel pole, put the tower's drop test in the order it has to happen in, and derive what a hydraulic ram multiplies. Marsh has the season on a wall chart with 15 working days left on it.

BRIEFING 15W

The carousel's poles and the flume's gate both hold, and neither has ever been computed.

WHAT YOU DECIDE 11W

Put forces on the parts that hold people in a circle.

5.1 What a pole holds sideways

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · BALLPARK

SCENE 34W

An outer horse carries a 78 kilogram rider 6.2 metres from the axis. The carousel is turning at eight revolutions a minute. Bisi Adeyemi wants the sideways force before anyone raises the controller setting.

GUIDE 65W

Five numbers, and two of them belong to earlier or other steps: the turns a minute, and gravity. Ask of each whether this force depends on it. Turns a minute is not a speed, so one number here is already the conversion done for you. And note the square: nine turns instead of eight is twelve per cent more speed and a quarter more force.

ASK 18W

Estimate the sideways force on the rider, converting the carousel's eight revolutions a minute into linear speed first.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a body in a circle needs a net force toward the centre · that force is the mass times the centripetal acceleration

VERDICT 76W

Eight revolutions a minute is an angular rate, not the rider's linear speed. One revolution covers $2\pi r$, so the rider moves about 5.2 m/s at 6.2 m radius. Then $F = mv^2/r$ gives about 340 N toward the centre. The square on speed makes controller changes expensive. Going from eight to nine rpm raises speed by 12.5%, but the required inward force by about 27%. The pole and saddle must supply that extra force every turn.

TAKEAWAY 20W

Circular-force problems often hide the needed speed inside a rotation rate, so convert the motion before applying $F = mv^2/r$.

5.2 The order a drop test runs in

THE DROP TOWER · DROP TOWER CONTROL · A PERSON · SEQUENCE

SCENE 39W

The safety engineer must observe and sign off on the tower's annual drop test. The tower technician has the test weights, the accelerometer and four operations on a whiteboard. They are written in the order he thought of them.

GUIDE 61W

All four operations will happen, so ask what each one has to be true for. A measured acceleration belongs to a stated mass, so the mass has to be recorded first. An instrument fitted after the drop measures nothing. And the brake only meets its real entry speed from full height. The last card is what turns a demonstration into evidence.

ASK 8W

Order the operations for the official drop test.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a test is only evidence if the conditions are known before it runs · test weights can stand in for passengers

VERDICT 74W

An official test only means something if its conditions are fixed before release. First record the test mass, because the measured acceleration belongs to that mass. Then fit and confirm the accelerometer before anything moves. The carriage must be released from full height so the brake meets its normal operating entry speed. Only after the drop can the logged result be compared with the certificate limit. The final comparison turns a demonstration into evidence.

TAKEAWAY 12W

A test is designed backwards from the number it has to produce.

5.3 Two centimetres driving nine

THE LOG FLUME · FLUME PUMPHOUSE · A RIDE · DERIVE

SCENE 33W

The flume's sluice gate is held by a hydraulic ram. A two-centimetre piston at the hand pump drives a nine-centimetre piston at the gate. An old note beside the drawing says only 'plenty.'

GUIDE 50W

Build Pascal's principle down to the force at the gate, one line at a time. Then state the cost: a hydraulic system multiplies force and gives something else away, and the derivation shows you exactly what, which is more use than the note on the drawing that says only 'plenty'.

ASK 16W

Work Pascal's principle down to the force at the gate, and say what the multiplication costs.

BEHIND THE BUTTON 168W

What the principle says. Pressure applied to a confined fluid is the same everywhere in it, so the small piston and the large one see the same pressure. Force is pressure times area, and area goes as the square of the diameter — which is why a two-centimetre piston driving a nine-centimetre one multiplies the hand force by about twenty.

What is given up. Volume is conserved: the fluid pushed out of the small cylinder is the fluid arriving in the large one. So the gate piston moves as much less distance as it gains force, and the work in equals the work out. Nothing has been created, and the pump handle has to travel a long way.

Why 'plenty' is not an answer. A number is what tells you whether the ram closes the gate against a head of water, how many strokes it takes, and what happens as the seals wear. The old note was true and useless, which is the difference this stop is about.

TAKES AS READ

pressure applied to an enclosed fluid is passed on undiminished · pressure is force divided by the area it acts over

VERDICT 76W

Pascal's principle sends the same pressure through the enclosed fluid. Force therefore scales with piston area, not with piston diameter. A diameter ratio of 9 to 2 gives an area ratio of about 20, so 250 N becomes roughly 5.1 kN. The gain in force is paid for in distance. The large piston moves about one twentieth as far as the small one. Work is conserved apart from losses, so the hydraulic creates no free energy.

TAKEAWAY 13W

An ideal machine can trade force for distance, but it cannot create work.

5.4 60 people and the same 5 seconds

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A RIDE · CALLBACK · CHOICE

SCENE 42W

Sam Idowu has run the pirate ship for 20 years. During low-angle maintenance swings, a full boat and an empty one both take 5.9 seconds. He wants to know why loading changes the force on the pivot bearings but not that timing.

GUIDE 68W

All four options explain a period that does not move when the boat fills. Ask of each whether it names a cancellation or an accident. The restoring pull is the boat's own weight, and the inertia resisting it grows in the same proportion. So they cancel, and length and gravity are what remain. Note the limit too: this holds at small angles, and the ride swings 48 degrees.

ASK 10W

Why does adding riders leave the low-angle period essentially unchanged?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a pendulum has a restoring force toward its lowest point · Newton's second law connects net force, mass and acceleration

VERDICT 76W

The restoring force comes from the boat's own weight. Add mass and that force grows in the same proportion as the inertia resisting acceleration. Those factors cancel, so the small-angle period does not depend on mass. Length and local gravity remain. This statement has a limit: the simple period formula assumes a small swing angle. At the ride's 48-degree operating swing, the period is a little longer. Loading still barely changes it, but amplitude now matters.

TAKEAWAY 21W

> A variable can disappear from a motion when it scales both the restoring force and the inertia in the same way.

END OF THE DAY 15W

Every part in this park is holding a force somebody can compute in three lines.

DAY 6 OF 15

What the collision keeps

BEFORE THE DAY — HUNT

Seven restraint sensors, and five are on the test rig

The certificate needs every restraint sensor tested, and two are not where the sheet says — they came off during the winter work. A sensor nobody can find is a restraint nobody can prove, and the ride does not open on the ones that were tested.

PLAN CARD 117W

Monday of the second week, and Wei Chen has 400 collisions off last season's accelerometers with nothing to compare them against. The bumper floor is the ride Vey expects to sign first and Georgia Hart expects to argue about, because whatever number comes out of it has to be a rule 12 nineteen-year-olds can apply on the four-hundredth cycle of a hot Saturday. Today you derive the speed a pair of cars leaves a collision at, work out what the lift motor is doing on the coaster, and settle what makes a wheel hard to start. Marsh has moved the reopening announcement to Friday, which gives the second week a deadline it did not have this morning.

BRIEFING 12W

400 logged collisions, and nobody has ever written the equation for 1.

WHAT YOU DECIDE 15W

Get the speed out of a collision, and the energy that does not come back.

6.1 Two cars, one velocity

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · DERIVE

SCENE 31W

A 320 kilogram car and rider at 3.1 metres a second strikes a stationary 290 kilogram car and rider. They stay in contact and move off together, which the telemetry confirms.

GUIDE 53W

Build the momentum statement down to the speed the pair leaves at, one line at a time. Then answer the second half: work out what fraction of the kinetic energy is still there afterwards. Momentum and energy behave differently in this collision, and the derivation is how you find out that they do.

ASK 20W

Work the momentum statement down to the speed the pair leaves at, and then say what happened to the energy.

BEHIND THE BUTTON 165W

Why momentum is the one that survives. In the moment of contact the two cars push on each other equally and oppositely, so whatever momentum one gains the other loses and the total is unchanged. That holds whatever happens inside the bumpers, which is why the momentum statement can be written before knowing anything about them.

Why the energy is not conserved. The cars stay in contact and move off together, which means the collision is perfectly inelastic — the maximum energy that can go into deformation, heat and noise for the given masses. The fraction remaining follows from the masses alone, and it is a fixed number rather than a measurement.

What the ride is doing with it. A bumper car is designed to lose that energy in the rubber rather than in the rider's neck, so the missing fraction is the whole safety argument of the attraction. Telemetry that confirms the pair moved off together is confirming which case of the algebra applies.

TAKES AS READ

momentum is mass times velocity, counted with its direction · a collision with no outside push conserves total momentum

VERDICT 71W

With negligible external impulse, total momentum is conserved during the collision. Because the cars leave together, one shared speed describes the pair afterward. Solving the momentum equation gives about 1.63 m/s. Kinetic energy is different. It falls because this collision is inelastic. The missing mechanical energy becomes deformation, sound and heat in the bumpers and bodies. Soft structures are useful precisely because they absorb energy instead of returning it as rebound.

TAKEAWAY 20W

For a perfectly inelastic collision, momentum fixes the shared speed while kinetic energy tells how much mechanical energy was transformed.

6.2 What the chain is paid to do

THE COASTER · COASTER STATION · A PERSON · BALLPARK

SCENE 34W

The chain hauls a loaded train of 4,800 kilograms up to 26 metres in 52 seconds. Kovač has the motor's plate rating and wants to know how much of it the lift actually uses.

GUIDE 68W

Five numbers, and one of them is the speed at the foot of the drop, which belongs to a different question. Ask of each whether the lift's work depends on it. Power is energy each second, so the answer needs both the energy stored and the time taken. Note what follows: nothing adds energy after the top, so every later hill fits inside what the lift paid for.

ASK 10W

Estimate the power the lift chain delivers to the train.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

lifting a mass stores energy equal to its weight times the height · power is work divided by the time taken

VERDICT 77W

Power is energy transferred each second. Raising the 4,800 kg train by 26 m stores about 1.22 MJ of gravitational potential energy. Doing that in 52 s requires about 23.5 kW delivered to the train. A faster lift would need more power for the same height. Once the train leaves the top of the first hill, no motor adds energy around the circuit. Friction only subtracts, so every later hill must fit inside what the lift supplied.

TAKEAWAY 17W

A motor is rated in power, and what a job needs is an energy and a time.

6.3 Where the mass is

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · CHOICE

SCENE 37W

Novák is quoting a replacement drive and the supplier wants the wheel's rotational inertia. Two wheels of the same total mass are on the table, one with heavy passenger cabins and one with a heavier center hub.

GUIDE 64W

Two wheels of the same total mass, and four options. Ask of each whether it appeals to where the mass sits, or only to how much there is. Each piece counts in proportion to the square of its distance from the axis. So a kilogram at the rim is worth many at the hub. Two of these options get the direction of that backwards.

ASK 15W

Two wheels have the same total mass. Which is harder to bring up to speed?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a torque is needed to change how fast something turns · the same mass can be arranged near an axle or far from it

VERDICT 76W

Rotational inertia depends on both mass and where that mass sits. Each piece contributes in proportion to the square of its distance from the axis. A kilogram near the outer rim matters far more than a kilogram near the center hub. That makes outer-rim-heavy wheels harder to speed up or slow down. Ferris-wheel passenger cabins are therefore kept light when possible. A flywheel does the opposite on purpose, putting mass outward so its rotation resists changes.

TAKEAWAY 14W

Rotational inertia counts every kilogram by the square of how far out it sits.

6.4 9 metres to take back 29

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · CALLBACK · BALLPARK

SCENE 33W

The carriage reaches the brakes at 28.7 metres a second and stops 9 metres later. The tower technician wants that turned into an acceleration before anyone argues about whether the stop feels acceptable.

GUIDE 59W

Five numbers, and two of them belong to the fall rather than the stop: the free-fall height above, and gravity. Ask of each whether the braking depends on it. And note the square on the entry speed. Halve the stopping distance and the average deceleration doubles, which is why a small rise in entry speed costs so much brake.

ASK 8W

Estimate the average deceleration in the braked section.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the same constant-acceleration relation runs backwards for slowing down · an acceleration can be quoted as a multiple of g

VERDICT 70W

The stopping relation is the same constant-acceleration equation used on the drop, but solved for a instead of v. With 28.7 m/s removed over 9.0 m, the average deceleration is about 45.8 m/s^2, or 4.7 g. The square on speed is the design lesson. Halving the stopping distance doubles the average deceleration. Likewise, a modest increase in entry speed demands much more braking force and leaves less room for error.

TAKEAWAY 18W

A stopping distance and a speed are enough to fix the acceleration, with no time measured at all.

END OF THE DAY 16W

A collision hands back its momentum in full and keeps a good share of its energy.

The period that ignores the load

PLAN CARD 118W

Tuesday, and Sam Idowu has been saying the same thing for two seasons: on a heavy boat the second push arrives late and you can feel it through the floor. He has been told each time that the ride is within specification, which it is. Today you derive the ship's period from the restoring force, so that there is a number to check the drive against rather than an operator's account of a floor. You also float a loaded boat at the flume and settle what the tower's timing gives for the local strength of gravity. Elena Bianchi has the swing arm's welds booked for Thursday and wants to know what load she is looking for cracks against.

BRIEFING 18W

Idowu has said for two seasons that the drive pushes at the wrong moment on a full boat.

WHAT YOU DECIDE 13W

Derive why a pendulum's timing does not depend on what is in it.

7.1 Why the mass falls out

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A RIDE · DERIVE

SCENE 37W

The boat hangs 8.6 metres below its pivot. In a low-angle maintenance test, Idowu logs 5.9 seconds per cycle whether the boat is full or empty. Nobody at the park can produce the expression that predicts it.

GUIDE 45W

Build the small-angle period from the restoring force, a line at a time. Then finish the job properly and state where the small-angle result stops being exact. Idowu's 5.9 seconds was logged in a low-angle test, and the ride does not run at low angles.

ASK 19W

Derive the small-angle period from the restoring force, then state when that result should not be treated as exact.

TAKES AS READ

for a small angle, the sine of the angle is close to the angle itself in radians · motion with a restoring force proportional to displacement repeats at a fixed rate

VERDICT 74W

For a small angle, $\sin\theta$ is close to θ, so the restoring force becomes proportional to displacement. That is the condition for simple harmonic motion. Mass appears in both the restoring force and ma, then cancels. The result is $T = 2\pi\sqrt{(L/g)}$, which gives about 5.9 s for 8.6 m. The ride operates near 48 degrees, well outside the small-angle range. Its operating period must therefore be measured rather than assumed from this formula.

TAKEAWAY 16W

An approximation can preserve the right physical dependencies while becoming quantitatively wrong outside its valid range.

7.2 How far a full boat sits down

THE LOG FLUME · FLUME PUMPHOUSE · A PERSON · BALLPARK

SCENE 36W

Fiona McCarthy loads the boats by eye and still follows an inherited four-adult rule. The moulded hulls replaced the timber boats two seasons ago. Each new hull is 2.9 metres by 1.1 and weighs 180 kilograms.

GUIDE 63W

Five numbers, and one of them is the depth of the hull side, which belongs to the freeboard question rather than to this one. Ask of each whether the sinking depends on it. A boat settles until it has pushed aside its own mass of water. So the answer needs a mass and an area, and the side height decides what happens next.

ASK 12W

Estimate how deep a boat with four adults sits in the water.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a floating body pushes aside its own weight of water · water has a density of about 1000 kilograms per cubic metre

VERDICT 82W

A floating boat settles until the buoyant force equals its weight. That means the displaced water must have the same mass as boat plus passengers. Four adults bring the total to 492 kg, which requires about 0.492 m^3 of displaced water. Spread over the hull area, that is about 0.154 m of depth in the water. A fifth adult still floats. The concern is the less hull side left above the water when the boat reaches splash-down and meets a rising wave.

TAKEAWAY 15W

A floating body settles until the water it has pushed aside weighs what it does.

7.3 Measuring gravity with a boat

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · CHOICE

SCENE 35W

The ship's low-angle maintenance period is 5.90 seconds with an 8.60 metre effective length. The safety engineer notes that the tower calculations use a local value of g nobody at this park has measured directly.

GUIDE 58W

Four options, and two of them agree on the number and disagree on where it applies. Ask of each what the measurement is a property of: the ship, or the place. Length and time are what went in, and neither mentions the boat. One option gets the algebra wrong in a way that produces a plausible-looking small number.

ASK 16W

What value of g follows from the low-angle period, and why can the tower use it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the pendulum period depends on the length and on g alone · an expression can be rearranged to make any of its quantities the subject

VERDICT 70W

Rearrange the small-angle period relation to $g = 4\pi^2 L/T^2$. The inputs are a length and a time, both easy to measure over many cycles. Using 8.60 m and 5.90 s gives about 9.75 m/s^2. That value belongs to the location, not to the ship, so it can be used in the tower, coaster and flume calculations. The low-angle condition matters because the simple pendulum relation is the model being used.

TAKEAWAY 20W

A physical constant measured by one system can calibrate another when both use a valid model of the same place.

7.4 The number people get wrong

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · CALLBACK · CHOICE

SCENE 38W

Jan Novák has a six-metre crane arm raised 30 degrees above horizontal near the Ferris wheel. A load hangs vertically from its end. His lifting plan and the shop sketch disagree about the distance used to calculate torque.

GUIDE 63W

Four distances, and only one of them is perpendicular to the line the load pulls along. Ask of each which direction it runs in. The load hangs straight down, so the arm that matters runs horizontally. The steel is six metres long and not all of it counts. Raising the arm shortens the horizontal distance, so the same load twists the pivot less.

ASK 20W

The crane arm is 6 metres long and angled up at 30°. Which distance sets the torque about the pivot?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a force applied further from a pivot has a greater turning effect · a force can be drawn as a line extending in both directions

VERDICT 74W

Torque uses the perpendicular distance from the pivot to the line along which the force acts. The load hangs vertically, so the needed arm is horizontal. For a 6 m crane arm at 30 degrees, that arm is 6 cos30°, or about 5.2 m. The full steel length is not the lever arm. Raising the arm shortens the horizontal arm, so the same hanging load produces less torque about the pivot. That distinction matters.

TAKEAWAY 22W

> A turning effect is fixed by where the line of the force passes, not by the shape of the thing carrying it.

END OF THE DAY 16W

A measured period is a measurement of the pendulum, and it can be turned into g.

DAY 8 OF 15

The arm and the crack

PLAN CARD 115W

Wednesday, and Elena Bianchi has an indication 41 millimetres long in the weld at the root of the wheel's number-nine spider arm. She has photographed it, sized it, and refused three times to say whether it matters, which is her discipline rather than her opinion: a crack is a length in a material and whether it counts depends on the load, and the load is not her number. Priya Raman wants that load computed today. Today you derive the force in the hub bolt group from the out-of-balance moment, put a number on what a bumper collision does to a neck, and read the carousel's drive panel. Marsh has the ride on the reopening poster.

BRIEFING 16W

Bianchi found an indication in a spider arm weld and will not say whether it matters.

WHAT YOU DECIDE 12W

Get a load into a joint before anybody looks at the joint.

8.1 What one wheel-center bolt is holding

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · DERIVE

SCENE 36W

The loading imbalance reaches 177 kilonewton-metres. Each support arm joins the wheel center through eight bolts, each 0.42 metres from the center. Priya Raman points to number nine and asks what force one bolt must carry.

GUIDE 56W

Build the torque statement down to the force in one bolt, one line at a time. The last line is not a number: you have to state the load-sharing assumption you just used, because the arithmetic only holds if the eight bolts carry equal shares — and Raman is asking about a specific bolt, number nine.

ASK 15W

Use the torque to find the force in one bolt, and state the load-sharing assumption.

BEHIND THE BUTTON 156W

What the balance says. The applied moment has to be carried by the bolts' forces acting at their distance from the centre. Eight bolts on a circle of radius 0.42 metres therefore share the job, and each one's contribution is its force times that radius. Setting the total against 177 kilonewton-metres gives the force per bolt.

Why the assumption is the answer's fine print. Equal sharing needs the bolts equally tight, equally fitted and the joint stiff enough not to favour some over others. A slack or worn bolt sheds its share onto its neighbours, so the number you compute is a lower bound on what the worst bolt is really carrying.

Why number nine. It is the one nobody has a final answer about, which makes it the one where the assumption matters most: if the equal-share assumption fails anywhere it fails there, and the computed force is exactly the number the certification will quote.

TAKES AS READ

torques about a point add to zero when nothing is turning · a ring of identical bolts shares a moment equally between them

VERDICT 75W

Turning a torque into a force needs a distance to divide by. Choosing that distance is the engineering. Here it is the bolt radius, since that is where the resisting forces act. About 52.7 kilonewtons per bolt is now the load the crack must be judged against. The assumption under it is worth saying out loud: equal sharing is what a drawing assumes, and an eighty-one-year-old joint only approximates it. Real joints can redistribute load.

TAKEAWAY 14W

A moment becomes a force the moment somebody asks what distance is resisting it.

8.2 The tenth of a second that matters

THE BUMPER CARS · BUMPER CAR PAVILION · A PERSON · CHOICE

SCENE 34W

Wei Chen's accelerometers show the car changing speed in about 0.10 seconds and the rider's head in about 0.18. The safety lead wants to know which interval belongs in a rule about rider force.

GUIDE 62W

Four options, and two intervals on the trace. Ask of each whose momentum change the time belongs to. The car takes a tenth of a second. The rider takes longer, because padding and belt stretch let them arrive at the new speed gradually. Using the car's interval describes the car. That is also why safety design lengthens rather than shortens the time.

ASK 13W

Which time should the force on a rider be computed over, and why?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a change in momentum needs a force acting over a time · a body and the seat it sits in can change speed over different times

VERDICT 72W

Impulse connects force, time and momentum change. The correct time interval belongs to the body whose force you want. The rider's head and torso take about 0.18 s to match the car's new speed, longer than the bumper contact itself. Using 0.10 s would describe the car more closely and would overstate the rider force. Safety systems work by lengthening the rider's stopping time through padding, belt stretch and controlled seat motion.

TAKEAWAY 19W

> The same speed change over a longer time is a smaller force, and the time is a design choice.

8.3 The drive that pulls unevenly

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · DIAGNOSIS

SCENE 31W

The carousel drive has drawn uneven current since winter. Bisi Adeyemi has five readings from the control panel and the ring survey beside them. Two look alarming. Three are stubbornly ordinary.

GUIDE 56W

Five readings, and the three ordinary ones do the work. The temperature has not moved. The speed is exactly where it was set. The average current is normal. Ask of each candidate how many of the five it covers. Something that peaks once per revolution is something fixed on the ground being met by something turning.

ASK 5W

Which explanation fits every reading?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a drive works harder when the load on it is greater · a bearing that is failing takes more torque to turn

VERDICT 85W

Once per revolution is the reading that decides it: something fixed on the ground is being met by something turning, once a turn. A failing centre bearing would draw more current all the way round and would be warm, and the temperature has not moved. A motor fault would not synchronise itself to the platform's rotation. And the speed is exactly where it is set, so nothing is running fast. The 11 millimetres is small, and it is at the right place in the cycle.

TAKEAWAY 11W

The readings that are normal are what rule an explanation out.

8.4 What nine metres of water pushes with

THE LOG FLUME · FLUME PUMPHOUSE · A RIDE · CALLBACK · BALLPARK

SCENE 33W

The water surface in the elevated tank is 9.4 metres above its outlet. Kelechi Abara has a pressure gauge on that outlet and an old notebook value that does not match the reading.

GUIDE 65W

Five numbers, and two of them describe the tank rather than the water above the outlet: the diameter, and the flow through it. Ask of each whether pressure depends on it. A narrow pipe of the same height gives the same pressure, which is why the width is offered and not needed. The outlet answers to the depth above it, however big the tank looks.

ASK 10W

Estimate the pressure above atmospheric pressure at the tank outlet.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

pressure in a liquid grows with depth below the surface · water has a density of about 1000 kilograms per cubic metre · newton's second law as the definition of a net force — taken as read

VERDICT 77W

Hydrostatic pressure comes from the weight of liquid above each unit area. It depends on density, gravity and depth, not on tank width. For 9.4 m of water, the pressure above atmospheric pressure is about 92 kPa. A narrow vertical pipe of the same height would produce the same pressure. That is why tank diameter is a distractor here. The outlet hardware responds to the pressure at its depth, even when the tank around it looks enormous.

TAKEAWAY 15W

Depth sets pressure, and the width of the tank has nothing to do with it.

END OF THE DAY 16W

A defect cannot be judged until somebody has computed what the part it is in carries.

DAY 9 OF 15

The loop is not the loop on the drawing

BEFORE THE DAY — CANVASS

What the old crew remember about the ship

The swinging ship has a fault history that never made it into the log, and half the original crew still live in the town. Ask around until enough accounts agree on what it used to do at the top of the swing, because a complaint pattern is evidence and a rumour is not.

PLAN CARD 137W

Thursday, and Marta Kovač put a tape over the crown of the loop yesterday afternoon because nobody could find a reason not to. The drawing says the radius there is 5.6 metres. The tape says 7.4. In Brennan's ninth notebook, under 1998, there is a line about regrading the top of the loop after complaints of a rough ride, and no drawing was ever changed. Vey is right this morning and says so quietly: a ride that has run for 25 years on a number nobody checked is not the same thing as a ride with a margin. Today you derive the speed the loop demands at its crown, settle what a margin actually is, and read the ship's drive against yesterday's period. Everything computed this fortnight above the first drop now has to be done again.

BRIEFING 19W

Kovač taped the crown of the loop yesterday and the radius is not the 1 on the 1974 drawing.

WHAT YOU DECIDE 11W

Derive what the loop demands, and measure what it actually is.

9.1 What the loop asks for at the top

THE COASTER · COASTER STATION · A RIDE · DERIVE

SCENE 37W

At the top of the loop the track is above the train and the rider is upside down. Kovač's tape gives a radius of 7.4 metres there, against 5.6 on the drawing everyone has been working from.

GUIDE 63W

Build the force statement at the crown of the loop down to a minimum speed, one line at a time. Then evaluate it twice — once for the 7.4 metres Kovač taped and once for the 5.6 on the drawing — because the point of the derivation is that it lets you answer for a radius nobody measured when the ride was designed.

ASK 21W

Work the force statement at the top of the loop down to a minimum speed, and evaluate it for both radii.

BEHIND THE BUTTON 190W

Why the crown is the critical point. Upside down at the top, both the track's push and the rider's weight point the same way: towards the centre of the loop. That is the one place on the circuit where gravity is helping to turn the train, so it is the place where the least speed is needed to stay on the track — and below that speed the train does not fall off, the track has to pull, which it cannot.

What the minimum condition means. Setting the track force to zero is not a design target; it is the boundary of the possible. A real ride is engineered well above it, and the margin is the whole safety case. The derivation gives you the boundary, and the certificate gives you the margin.

Why two radii matter so much. The minimum speed goes as the square root of the radius, so a loop that is actually 7.4 metres across the crown demands more speed than a 5.6-metre one — and every calculation done from the drawing has been demanding too little. That is a paperwork problem with a physical consequence.

TAKES AS READ

a body moving in a circle needs a net force toward the centre of that circle · a rail can push a train but cannot pull it

VERDICT 78W

At the top of the loop, both gravity and the rail can point toward the centre. The limiting case occurs when the rail's normal force just falls to zero. Then gravity alone must supply mv^2/r, giving $v_{min} = \sqrt{gr}$. A larger radius at the top of the loop therefore requires a larger minimum speed. Using 7.4 m gives about 8.5 m/s, compared with 7.4 m/s from the old 5.6 m drawing. Smoother geometry can still demand more speed.

TAKEAWAY 17W

The limiting case is the one where a part stops doing anything, and it sets the number.

9.2 What a safety factor says

THE DROP TOWER · DROP TOWER CONTROL · A PERSON · CHOICE

SCENE 38W

The inspection file gives the coaster a margin of 1.38 using the old loop drawing. The measured top of the loop reduces it to 1.20. The park manager wants to know whether that still counts as a pass.

GUIDE 68W

Four readings of a margin that fell from 1.38 to 1.20. Ask of each whether it puts the change in the ride or in the knowledge of it. A margin is capacity divided by demand, and only one of those moved. Nothing about the train changed over the winter. And a smaller known margin is safer than a larger imaginary one, because it can drive an operating rule.

ASK 17W

The margin on the top of the loop falls from 1.38 to 1.20. What has actually changed?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a margin is a ratio between what is available and what is required · a requirement can be recomputed when the geometry it rests on changes

VERDICT 86W

A safety margin is capacity divided by demand. Here the train's available speed at the top of the loop has not suddenly changed. The measured radius at the top of the loop changed the required speed. That raises the demand and lowers the ratio from 1.38 to 1.20. The physical ride was already in this state before anyone measured it correctly. What changed today is the park's knowledge. A smaller known margin is safer than a larger imaginary one because it can drive an operating rule.

TAKEAWAY 16W

A margin is a ratio, and it means nothing until both of its numbers are named.

9.3 Idowu was right

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A RIDE · DIAGNOSIS

SCENE 35W

The drive pushes the boat at a fixed interval set years ago. A low-angle test gives 5.9 seconds, while the 48° operating swing measures about 6.15. Five console readings are available and three look ordinary.

GUIDE 63W

Five readings, and the ordinary ones matter. The bearing temperature is normal. The drive delivers its rated push. Ask of each candidate how many of the five it covers. The deciding reading is the operating period, which at 48 degrees is not the small-angle figure the drive was set to. Heavy loads expose a timing mismatch sooner, because they need more good pushes.

ASK 12W

Which explanation fits the full set of readings, including the operating period?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a push in step with an oscillation adds energy and an out-of-step push can remove it · the small-angle pendulum formula is approximate at a 48-degree swing

VERDICT 82W

The operating period is the deciding reading. At 48 degrees the small-angle estimate is no longer exact, and the measured cycle is about 6.15 s. The bearing temperature is normal, so extra damping from drag is not the main loss. The drive also delivers its rated push. The remaining problem is timing. Each early push reaches the boat at the wrong phase and transfers less useful energy. Heavy loads expose the mismatch sooner because they need more well-timed pushes to reach angle.

TAKEAWAY 16W

> Resonance depends on phase, so the useful period is the one the operating system actually follows.

9.4 **Going round at a steady speed**

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · CALLBACK · CHOICE

SCENE 33W

The carousel turns at eight revolutions a minute and has done so since the previous engineer set it. Bisi Adeyemi is wiring a new controller and wants the outer-horse acceleration put on paper.

GUIDE 63W

Four options, and two of them agree on the size and disagree on the direction. Ask what is changing when a speed stays constant. Velocity carries a direction as well as a size, and on a circle the direction changes every instant. So decide which way the acceleration points before worrying about its value. One option needs a mass the question never gives.

ASK 16W

An outer horse goes round at a constant eight revolutions a minute. What is its acceleration?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

velocity has a direction as well as a size · acceleration is the rate at which velocity changes · velocity against speed: direction is part of the quantity — taken as read

VERDICT 66W

Speed can stay constant while velocity changes because velocity also contains direction. On a circle, that direction changes every instant. The resulting acceleration points toward the centre and has size v^2/r. Here 5.2 m/s on a 6.2 m radius gives about 4.4 m/s^2. The platform and pole supply the inward force. A rider feels the seat pushing inward while their inertia makes the experience seem outward.

> **TAKEAWAY** 15W
>
> Acceleration describes any change in velocity, including a change in direction when speed stays constant.

END OF THE DAY 19W

A flatter loop is a gentler ride and a hungrier one, and both of those are the same number.

Everything above the first drop, again

PLAN CARD 130W

Friday, and the second week ends with the coaster's file open on the bench and every number in it suspect above the first drop. The train crosses the crown at about 10.2 metres a second and the crown now demands 8.5, which is a margin of 1.20 rather than the 1.38 the certificate claimed. Kovač's clipboard says the station return speed has fallen 0.9 metres a second in 3 seasons. Today you turn a collision into a force on a neck, decide what to change on the bumper floor, and size the pump that has to lift the flume's water back up. Vey will not sign the coaster on a margin that is shrinking unless somebody can say how fast it shrinks and what would be measured to catch it.

BRIEFING 12W

Every margin computed before yesterday rests on a radius that was wrong.

WHAT YOU DECIDE 9W

Recompute the circuit against the loop it actually has.

10.1 Turning a collision into a force

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · DERIVE

SCENE 33W

The rider's own speed change in Wei Chen's telemetry is 1.47 metres a second over 0.18 seconds. The safety lead wants that turned into a force and then compared with the rider's weight.

GUIDE 53W

Build the impulse statement down to an average force on a sixty-kilogram rider, one line at a time. Then express that force as a multiple of the rider's own weight. In newtons the number means nothing to the safety lead; as a multiple of body weight it is a comparison they can use.

ASK 21W

Work the impulse statement down to an average force on a 60 kilogram rider, and express it against their own weight.

TAKES AS READ

impulse is force times time and equals the change in momentum · a rider's weight is their mass times g

VERDICT 75W

Impulse equals the change in momentum, so average force is momentum change divided by contact time. The rider changes speed by 1.47 m/s over 0.18 s. For a 60 kg rider, that gives about 490 N. Compared with body weight, the force is roughly 0.83 g. The design lever is time. Stretching the same momentum change over a longer interval lowers the average force in direct proportion, which is why padding and compliant restraints help.

TAKEAWAY 19W

A force from a collision is a momentum change divided by the time somebody was given to make it.

10.2 A margin that moves

THE COASTER · COASTER STATION · A PERSON · CHOICE

SCENE 31W

The top-of-loop margin is 1.20. Marta Kovač's clipboard shows station return speed falling about 0.3 metres a second each season. Each spring measurement used the same wheel and the same method.

GUIDE 59W

Four candidate rules, and the quantity that matters is hard to measure every morning. Ask of each whether the crew could actually do it before opening, every day. Station return speed is easy and consistent, and the ride's own losses link it to the speed at the top. A yearly recalculation cannot catch a decline soon after it starts.

ASK 14W

What should the certificate require, given a margin that falls a little each season?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

friction losses grow as wheels and bearings wears · a margin can be watched by measuring something easier than the quantity itself

VERDICT 81W

The quantity that matters at the loop is speed at the top of the loop, but measuring it every morning would be difficult. Station return speed is easy to measure consistently. The two are linked by the ride's energy losses, so one can serve as a practical stand-in for the other. A daily minimum catches a decline soon after it begins. A yearly recalculation does not. The rule turns a hidden margin into a number the opening crew can actually watch.

TAKEAWAY 13W

A quantity nobody can measure daily is watched through one that can be.

10.3 What the water costs to put back

THE LOG FLUME · FLUME PUMPHOUSE · A RIDE · BALLPARK

SCENE 35W

The flume returns 0.42 cubic metres of water a second to an elevated tank 9.4 metres up. Abara's pump is rated at 55 kilowatts, and he has never known whether that leaves much spare capacity.

GUIDE 53W

Six numbers, and one of them is the pump's rated capacity, which the answer is compared against rather than built from. Ask of each whether the power depends on it. Efficiency divides rather than multiplies, so getting it the wrong way round makes the pump look comfortable. It is rated at 55 kilowatts.

ASK 10W

Estimate the electrical power the pump needs at 72% efficiency.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

lifting a mass stores energy equal to its weight times the height · a pump delivers less than it draws, by its efficiency

VERDICT 71W

The flow rate times density is the mass moved each second. Raising that water through 9.4 m gives useful power of about 38.7 kW. Because the pump is only 72% efficient, the electrical input must be larger: about 53.7 kW. Set that against the 55 kW rating and only about two per cent is left over. That is why heat, wear or a little extra lift height can trip the pump.

TAKEAWAY 19W

Power sizing needs the useful lifting power and the efficiency loss before anybody can say what is left over.

10.4 One g, all the way down

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · CALLBACK · HOLD

SCENE 42W

The tower's magnetic brake is set from the console and the car has to come down inside the deceleration band the certificate names. The car's mass changes with every load, and the brake does not know how many people are in it.

GUIDE 46W

Hold the deceleration inside the band on the console. The band narrows toward the bottom, because the last stretch is where a rider's neck takes whatever the brake does. The brake current is your control, and every change in load changes what the same current produces.

ASK 10W

Hold the deceleration inside the band all the way down.

BEHIND THE BUTTON 137W

Why the same brake gives a different ride. A net force divided by mass is the acceleration, so the same braking force decelerates a heavy car less than a light one. The brake current sets the force; the riders set the mass; the certificate is written about the acceleration, which is the ratio.

Why nothing here is a step. A magnetic brake's force depends on speed, so as the car slows the force falls and the deceleration falls with it. Holding a deceleration means raising the current continuously through the stop rather than setting it once.

Why the band tightens near the bottom. The car is slowest there, the brake is weakest there, and the rider's tolerance is unchanged. It is the one part of the drop where the margin has to be made rather than inherited.

TAKES AS READ

a force applied to a mass produces an acceleration · acceleration as the rate a velocity changes, whichever part of it changes — taken as read

VERDICT 139W

The certificate limits the deceleration. The console sets the braking force. Between them sits the mass in the car, and that changes on every cycle. Twelve adults and four children is a different vehicle from four adults. A current that gave a comfortable stop this morning gives a harsher one when the car is light. That is the second law as an operating rule rather than an equation. There is a second difficulty. A magnetic brake's force depends on how fast the car moves through the field, so it fades as the car slows. Hold the current steady and the deceleration decays away with it. The current has to be raised through the stop. And the band narrows at the bottom, where the brake is weakest and the rider's neck is no stronger than it was at the top.

TAKEAWAY 17W

The same force is a different ride, because what the rider feels is force divided by mass.

END OF THE DAY 18W

A margin that moves is a schedule, and a margin that is checked each morning is a rule.

The day the flume behaved

PLAN CARD 119W

Monday of the third week, and nothing is wrong. The flume ran all weekend on the new pump curve, the coaster's morning speed came in where Friday's rule says it should, and Vey spent the afternoon reading rather than asking. Kelechi Abara has wanted a derivation for the outlet speed since the first week and this is the first day anybody has had time to do one. Today you derive what the water leaves the chute at, settle what the carousel's controller does when the limit is reached, and put a number on the wind load a gondola carries. Marsh has started telling people the park opens in 11 days, which she has not said out loud since October.

BRIEFING 15W

Nothing broke over the weekend, which is the first time that has happened this month.

WHAT YOU DECIDE 12W

Close the books on the water while there is nothing going wrong.

11.1 The same square root, from a different law

THE LOG FLUME · FLUME PUMPHOUSE · A RIDE · DERIVE

SCENE 35W

The chute outlet is 4.2 metres below the surface of the water standing in the top pond. Abara has a figure from the manufacturer and no derivation, and both surfaces are open to the air.

GUIDE 57W

Build the energy statement for the flow down to the speed at the outlet, one line at a time. Then say what it has in common with the drop tower on day one — the same square root arrives from a different law, and noticing that is the point of putting the two stops in one campaign.

ASK 23W

Work the energy statement for the flow down to the outlet speed, and say what it has in common with the drop tower.

BEHIND THE BUTTON 164W

Why the pressures cancel. Both surfaces are open to the air, so the pressure term is the same at each end and drops out. What is left relates the height difference to the speed, and the result is the same $\sqrt{(2gh)}$ the falling carriage gave — because in both cases gravitational energy is being turned into motion with nothing else taking a share.

What is being assumed. Steady flow, no viscosity, and a pond surface large enough that its own level barely moves. Real chutes lose to friction and turbulence, so the derived speed is an upper bound, and the manufacturer's figure being lower is what you would expect rather than a contradiction.

Why the same answer twice matters. A student who derives $\sqrt{(2gh)}$ from kinematics and again from an energy statement about a fluid has met the same physics wearing two different notations. That recognition is worth more than either result, and it is why the flume is on the syllabus at all.

TAKES AS READ

for steady incompressible flow, continuity keeps area times speed constant · pressure, motion and height together account for the mechanical energy of a flowing liquid

VERDICT 73W

Write the fluid's mechanical energy per unit mass between the upper surface and the outlet. Both points are open to air, so atmospheric pressure cancels. Continuity says the broad pond surface moves much more slowly than water in the narrow chute, so its kinetic term is negligible. The 4.2 m height drop then gives $v = \sqrt{(2gh)}$. The tower reached the same expression through kinematics. Both cases turn gravitational potential energy into motion.

TAKEAWAY 20W

> A height drop can become speed in a falling object or in a flowing liquid because both are energy conversions.

11.2 A limit somebody can pass

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · CHOICE

SCENE 32W

The derived limit for the swings is ten turns a minute. Adeyemi has to wire what happens when the controller is asked for 11, and there are 4 ways to do it.

GUIDE 61W

Four ways to wire a limit, and all four record something. Ask of each whether the machine enforces the number or a person does. An alarm still leaves the choice with the operator. A log arrives after the exceedance. And rounded inputs are not a licence to add ten per cent, because the clearance that produced the limit has not moved.

ASK 16W

What should the controller do when the speed setting is pushed past ten turns a minute?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a limit that is not enforced by the machine can be exceeded by an operator · a ride can be made to refuse a setting rather than to warn about it · velocity against speed: direction is part of the quantity — taken as read · centripetal acceleration: v^2 over r, pointing at the centre — taken as read

VERDICT 77W

A calculated limit matters only if the machine enforces it. An alarm still lets an operator choose to exceed the value. A log records the exceedance after it has already happened. The controller should therefore refuse settings above the derived limit. That makes the physics part of the machine rather than a suggestion beside it.

Rounding the inputs does not justify adding an arbitrary ten percent, because the physical clearance that produced the limit has not changed.

TAKEAWAY 18W

A safety limit only works when the response to crossing it is defined before the limit is reached.

11.3 What the wind does to a passenger cabin

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · BALLPARK

SCENE 31W

A passenger cabin presents about 4.2 square metres side-on. Raman wants the force on 1 at the 22 metres a second the county's rule names, before anybody argues about the limit.

GUIDE 63W

Six numbers, and one of them is the number of cabins on the wheel, which belongs to a later total. Ask of each whether the force on one cabin depends on it. And note the square on the wind speed. Twenty-two to thirty metres a second is nearly twice the force, which is why a rule watches gusts rather than the daily mean.

ASK 13W

Estimate the wind force on 1 passenger cabin at 22 metres a second.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

moving air exerts a force on a surface that grows with the square of its speed · air has a density of about 1.2 kilograms per cubic metre

VERDICT 73W

Wind force scales with the square of speed. For this passenger cabin, 22 m/s produces about 1.46 kN of sideways force. Doubling wind speed would quadruple that load. Even a rise from 22 to 30 m/s increases the force by about 86%. That steep growth is why gust speed matters more than the daily mean. A sensible operating rule starts action below the structural limit because unloading and stopping the wheel take time.

TAKEAWAY 22W

A wind limit is a force limit in disguise, and the square is why the last few metres a second matter most.

11.4 Angle and charge, at three marks

THE LOG FLUME · FLUME PUMPHOUSE · A PERSON · LOB

SCENE 34W

The old midway stall has a water cannon and three targets across the pond. The pressure dial is unmarked, the pump is thirty years old, and nobody has a number for what it delivers.

GUIDE 49W

Set the angle and the charge, and fire. Three marks, working outward. The dial is relative and the pump is worn, so nothing here tells you the launch speed — the way to find it is to fire, watch, and put the next shot between two you have seen.

ASK 8W

Land the jet on each mark, working outward.

BEHIND THE BUTTON 155W

Why there is no calculation to do first. Every other stop in this park hands you a number and asks what follows from it. This one hands you a dial with no units on a pump nobody has tested, which is the honest situation and the reason a fairground game is learned rather than solved.

What the flight tells you anyway. Once the water leaves the nozzle, gravity is the only thing acting on it downward, and it pulls the same whatever the jet weighs. Time in the air is set by the vertical part of the launch and nothing else, so a steeper shot hangs longer and lands shorter at the same charge.

Why past forty-five degrees it gets worse. Aim higher and more of the same charge goes into height instead of distance. The far mark wants a flatter shot than the near one, which is the opposite of what everybody tries first.

TAKES AS READ

a thrown or fired thing rises and falls under gravity

VERDICT 165W

There is no number to start from, and that is the point of this stall. The dial is unmarked, the pump is worn, and the only honest method is bracketing: fire short, fire long, and put the third shot between two settings you have actually watched. What the flights teach while you do it is the physics the rest of the park is built on. Once the water is in the air, gravity acts downward and nothing else does, and it pulls at the same rate whatever the jet weighs — so the time in the air is set by the vertical part of the launch alone. That is why a steeper shot hangs longer and still lands shorter at the same charge, and why past forty-five degrees the extra effort goes into height rather than distance. The far mark wants a flatter shot, which is the opposite of what everybody tries, and finding that out by firing is the fastest anybody ever learns it.

TAKEAWAY 15W

Some quantities are read off an instrument, and a few are found by trying them.

11.5 The lap that does not come back

THE COASTER · COASTER STATION · A RIDE · CALLBACK · DERIVE

SCENE 41W

The train leaves the top of the first hill at 26 metres and reaches the bottom of the first drop 2 metres above the station. Kovač's wheel says 21.4 metres a second at the bottom, and an ideal circuit says more.

GUIDE 65W

You build the derivation a line at a time, choosing at each step the expression the line above gives you. Every wrong branch is legal algebra, so the malformed-line trick will not help. Work the energy statement down to the speed at the foot of the drop with no losses — and then say what Kovač's measured 21.4 metres a second implies about the difference.

ASK 23W

Work the energy statement down to the speed at the foot of the drop, and say what the measured speed then tells you.

BEHIND THE BUTTON 178W

What the starting statement says. Energy at the crest equals energy at the bottom plus whatever friction took. Written out, that is height and speed at the top against height and speed at the bottom, with one term for the losses. Nothing about the shape of the track appears, which is why energy is the tool of choice here rather than forces.

Why the mass cancels, and when it does not. Every term in the ideal statement carries m, so it divides out and the ideal speed is independent of how loaded the train is. The friction term is not proportional to m in the same way, so the moment losses enter, the loaded and empty trains stop agreeing.

What the gap is worth. The difference between the ideal speed and 21.4 is the energy the circuit lost on the way down, expressed as a speed. Turning that into joules and then into a percentage is how a maintenance decision gets made: a lap that does not come back is a wheel, a bearing or a brake dragging.

TAKES AS READ

stored energy is weight times height, and kinetic energy is half the mass times the speed squared · friction does negative work along the whole path

VERDICT 80W

Start with mechanical energy at the top of the first hill and at the foot of the drop. Mass appears in every term and divides out. That is why loading the train does not change the ideal speed from the same height. The ideal result is about 21.8 m/s, while the measured value is 21.4. The difference is energy lost to friction. That loss can be tracked around the circuit, especially near the loop where the speed margin is smallest.

TAKEAWAY 16W

> The gap between the ideal speed and the measured one is a measurement of the friction.

END OF THE DAY 18W

The morning to close a set of books is the one where nothing is asking to be fixed.

DAY 12 OF 15

The first hot Saturday

BEFORE THE DAY — EVADE

The owner who wants an opening date today

He has been on the midway since eight wanting a date for the press, and Vey, the county ride inspector, has said plainly that the date comes out of the inspection rather than the other way round. Keep clear of him and get your round done.

PLAN CARD 113W

Tuesday, and the forecast for opening weekend came in this morning: 30 degrees and a westerly gusting through the afternoon. Georgia Hart's point has been the same all fortnight and today it is the whole agenda — whatever gets decided in this workshop has to be doable by a nineteen-year-old on the four-hundredth cycle of a hot day. Today you set the wind rules for the wheel with the lead time built into them, decide what to do about the ship's drive, and settle how the tower's brake gets proved rather than asserted. Vey has said he will witness whatever tests are run on Thursday and sign or not sign on Friday morning.

BRIEFING 15W

30 degrees forecast, a gusting westerly, and a queue that will not stop all afternoon.

WHAT YOU DECIDE 15W

Set the limits the ride will be operated to when nobody has time to think.

12.1 Stopping a wheel takes time

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A RIDE · TRIGGER

SCENE 35W

Emptying the wheel takes about eight minutes and clearing the queue takes longer. Raman has the forecast updates on the board and wants the queue threshold set before the gusts arrive rather than during them.

GUIDE 68W

One rule, written before the forecast updates arrive and fixed once released. Closing the queue and clearing the people already standing in it takes two hours, and at 22 m/s the wheel must not be carrying anybody. So the line goes back from 22 by however far the gusts climb in two hours — and not so far back that the ride is shut on a breezy afternoon.

ASK 19W

At what gust speed do you close the queue, given clearing the people already in it takes two hours?

BEHIND THE BUTTON 157W

Why a threshold is not a limit. The structural limit is where the wheel must already be stopped. The threshold is where you start stopping, and the gap between them is the time the action needs. Setting the threshold at the limit means beginning the evacuation at the moment it should have finished.

Why the two thresholds differ. Emptying the wheel and clearing the queue are different jobs with different durations and different consequences if they overrun. Two actions with the same lead time would be one rule; the reason there are two numbers to set is that one of them takes longer than the other.

Why the board is released rather than adjusted. Gusts arrive as a series of forecast updates, and adjusting a rule while the updates come in is how an operator ends up justifying whatever they have already done. Writing both numbers first is what makes them a rule instead of a commentary.

TAKES AS READ

a wheel cannot be emptied instantly, and every action here needs warning · wind force grows with the square of the speed

VERDICT 94W

A wind threshold is useful only if the action behind it can finish before conditions become unsafe. Those two jobs take an eight-minute unload and rather more than that to run the queue down, and both have to be over before the gusts arrive. Their lead times force the first triggers well below the 22 m/s structural limit. Wind force is also rising as speed squared during the same interval. Waiting until the limit is nearly reached leaves too little time. The rule must be written before the forecast stream arrives, not improvised afterward.

TAKEAWAY 14W

A threshold has to fire early enough for the action behind it to finish.

12.2 Three tenths of a second

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A PERSON · CHOICE

SCENE 33W

The drive pushes every 5.85 seconds while the 48° operating swing measures about 6.15. Sam Idowu, the structural engineer and the manufacturer's manual disagree, and the ride is already on the reopening poster.

GUIDE 59W

Four options, and three of them change the machine to preserve a setting. Ask of each whether it acts on the period or on the drive. The period comes from geometry, gravity and amplitude. The interval is a number somebody typed. Pushing harder at the wrong phase adds load without fixing anything, and mass mostly cancels from the period.

ASK 17W

The drive and the operating swing disagree by three tenths of a second. What should be changed?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a push in step with an oscillation adds energy and an out-of-step push can remove it · the operating period can differ from the small-angle estimate at large amplitude

VERDICT 82W

The operating period is set by the ride's geometry, gravity and amplitude. The drive interval is a control setting. Changing the arm would rebuild the machine to preserve an old setting. Pushing harder at the wrong phase adds structural load without fixing the timing. Restricting rider mass also misses the cause because mass largely cancels from the period. The direct fix is to retime the drive to the measured operating period and then verify that the swing reaches angle without excess load.

TAKEAWAY 11W

Resonance is a timing problem before it is a force problem.

12.3 What the drop test has to show

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · VERIFY

SCENE 36W

Thursday's official drop test has one afternoon to produce a number the safety engineer can compare with the certificate. The tower technician has test weights, a recording accelerometer and the full tower before the inspectors leave.

GUIDE 59W

Lock a prediction for the peak deceleration before the drop happens: the panel will not let you look first. Then run the test and read what the accelerometer actually recorded. The last step costs something — you have to say what the comparison settles, and a measurement you interpret after seeing it can be made to agree with anything.

ASK 17W

Predict the peak deceleration the brakes will produce, run the drop, and say what the measurement settles.

BEHIND THE BUTTON 162W

Why the prediction is locked. A number written down afterwards is not a prediction, and the way it fails is not dishonesty but memory: with a recording in front of you it is genuinely hard to recall how confident you were. Committing first is a procedure that protects you from yourself, which is why it is enforced by the panel rather than recommended.

What the test can and cannot settle. It can tell you whether the brakes produce the deceleration the certificate claims, on this tower, with these weights, today. It cannot tell you what a loaded carriage of nervous people will do differently, and the inspectors will ask about the difference.

Why one afternoon is the constraint. The inspectors leave, so a test that produces an ambiguous result is a test that cannot be repeated in time. That is the reason for deciding beforehand what result would count as agreement — the alternative is discovering the criterion after the data arrives.

TAKES AS READ

a prediction made before a test is worth more than an explanation offered after it · an instrument records what happens at the place it is fitted

VERDICT 79W

Day two gave the average deceleration from the speed and the distance. It is not the number a rider meets. A magnetic brake makes force in proportion to speed, so it bites hardest at the entry and eases as the carriage slows. The peak is therefore above the average, and the only way to know by how much is to fit an instrument and drop it. Predicting first is what makes the drop a measurement rather than a demonstration.

TAKEAWAY 15W

A test is only evidence about a claim that was written down before it ran.

12.4 100 collisions an hour

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · CALLBACK · CHOICE

SCENE 36W

Wei Chen has accelerometers in three bumper cars and 400 impacts logged from last season. In one clean test, a moving car hits a stationary car and the pair travel together for an instant after contact.

GUIDE 58W

Four options, and each keeps or discards two quantities. Ask of each what the collision does to the outside world in a tenth of a second. Almost no outside push arrives, which protects one of the two. The cars leave together, deformed and warm, which does not protect the other. That difference is what soft bumpers are for.

ASK 21W

A 320 kg car at 3.1 m/s strikes a stationary 290 kg car and they move off together. What is conserved?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

momentum is mass times velocity, counted with direction · kinetic energy depends on the square of the speed · velocity against speed: direction is part of the quantity — taken as read · newton's second law as the definition of a net force — taken as read

VERDICT 68W

Total momentum stays the same because the pair receives almost no external impulse during the brief collision. Kinetic energy does not. The cars leave together at about 1.63 m/s, carrying roughly 810 J of kinetic energy after starting with about 1,540 J. The missing mechanical energy became deformation, sound and heat. That

> distinction separates an inelastic collision from an elastic one and explains why soft bumpers reduce rebound.
>
> **TAKEAWAY** 15W
>
> A collision can preserve one bookkeeping quantity while transforming another into heat, sound and deformation.

END OF THE DAY 22W

A limit written for a calm Tuesday is not a limit; it has to work on the worst afternoon of the year.

Three things and one crew

PLAN CARD 124W

Wednesday, and three things arrived this morning: a bumper car with a cracked chassis rail, the coaster's morning speed a tenth low for the second day running, and the carousel's platform settling that Adeyemi first saw as a current peak. Kovač has one crew and two days. Marsh wants the coaster because it is on the poster; Hart wants the bumper car because it is the ride her staff cannot supervise their way out of. Today you rank the three, decide what the coaster's low reading means, and put the carousel's repair in an order that can be done overnight. Vey inspects on Friday morning and signs on Friday afternoon, and what is not fixed by Thursday night is a ride that opens shut.

BRIEFING 11W

Three faults, one maintenance crew, and the inspection is on Friday.

WHAT YOU DECIDE 8W

Rank what gets fixed with two days left.

13.1 Which of the three cannot wait

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · DELEGATE

SCENE 30W

A cracked chassis rail on 1 car, a coaster reading a tenth low, a carousel platform 11 millimetres down. One crew, two days, and Kovač wants the order in writing.

GUIDE 56W

Three faults, one crew, two days. Keep the one that needs your own judgement, and for each of the others choose an owner, the first thing they will do, and the reading or event that brings it back to you. Compare the three on trend, containment and consequence rather than on how bad each one sounds.

ASK 23W

Keep one problem yourself. Assign the other two so the crew knows what to do first and when to bring each one back.

BEHIND THE BUTTON 153W

What the three axes are. Trend is whether it is getting worse while you decide. Containment is whether the fault is already isolated from the public. Consequence is what happens if it is not dealt with. A cracked chassis rail, a coaster reading a tenth low and a platform eleven millimetres down score very differently on each.

Why the loudest is not the first. A fault that is already contained — the car taken out of service — is not getting worse, whatever its consequence would have been. A drift of a tenth in a reading is small now and is evidence of something that may not be small next week.

Why the order goes in writing. Kovač asks for it because a verbal priority list is one the crew re-derives at every interruption. An order with an owner, a first action and a return condition per item survives the day without you.

TAKES AS READ

a fault can be contained by removing only the affected equipment · a passing limit and a worsening trend are different kinds of information

VERDICT 81W

The cracked bumper car is serious but contained once that car is removed from service. The coaster is still above its morning limit, so today's reading belongs in the trend log. The carousel is different. Its foundation is moving under a ride that cannot isolate one section, and each cycle keeps loading the same low bearing. That makes it the rising, hard-to-reverse problem. Good delegation separates the problem you must stop now from problems that can return on a clear condition.

TAKEAWAY 19W

Urgency depends on how a fault changes and whether it can be isolated, not on how dramatic it looks.

13.2 A tenth of a metre a second

THE COASTER · COASTER STATION · A PERSON · CHOICE

SCENE 33W

The new morning rule sets a minimum station return speed. Today's reading is a tenth of a metre a second below yesterday's and still above the minimum, on a cold morning after rain.

GUIDE 62W

Four options, and the reading is below yesterday's and above the minimum. Ask of each whether it answers the limit question or the trend question. A limit says whether the ride may run today. A trend needs several mornings. One cold wet morning can move rolling resistance without proving wear. Two of these options make the rule unusable or the trend invisible.

ASK 14W

The reading is down a tenth and still passing. What does the rule require?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the morning reading stands in for a speed at the top of the loop nobody measures · rolling resistance changes with temperature and with wet track

VERDICT 72W

A limit and a trend answer different questions. Today's reading is still above the operating minimum, so the rule says the coaster may run. The lower value still belongs in the log. One cold, wet morning can shift rolling resistance without proving long-term wear. A sequence of similar readings would be different evidence. Stopping every passing ride makes the rule unusable, while explaining away an unrecorded reading makes a real trend invisible.

TAKEAWAY 15W

A rule with a limit needs a way of telling a trend from a Tuesday.

13.3 Leveling the carousel overnight

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · SEQUENCE

SCENE 35W

The carousel platform sits 11 millimetres low at one bearing. Marta Kovač has one night to lift the ring, add thin spacer plates under the support and prove the repair before the morning opening check.

GUIDE 57W

All four operations will happen tonight, so ask what each needs first. A bearing cannot be adjusted while the ring is pressing down on it. Spacer thickness comes from the measured level rather than from counting plates. And the last card is the test: if the once-a-revolution peak has gone, the measurement that started this is answered.

ASK 8W

Order the four operations of the overnight repair.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a load has to be taken off a bearing before its support height can be adjusted · a repair is finished when the original problem is measured again

VERDICT 72W

The repair order follows the load path. The low bearing cannot be adjusted while the ring is pressing on it, so nearby jacks take the weight first. Spacer thickness comes from the measured level, not from counting plates. After lowering the ring, the mounting bolts are tightened to the specified torque. The final run is essential. If the once-per-revolution current peak disappears, the measurement that started the repair has also been resolved.

TAKEAWAY 15W

A repair procedure is a chain of physical dependencies, not just a list of tasks.

13.4 What the swing is worth at the bottom

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A RIDE · CALLBACK · BALLPARK

SCENE 33W

The pirate ship reaches 48 degrees on a full cycle. Idowu wants the speed through the bottom before the bearing work is quoted. The loaded boat's centre sits 8.6 metres below the pivot.

GUIDE 61W

Five numbers, and two of them belong to other questions: the pivot-to-centre length, and the period. Ask of each whether the speed at the bottom depends on it. Energy answers to the change in height and not to the shape of the path. So the height risen is the term that matters, and the length is how it was worked out.

ASK 12W

Estimate the speed of the boat through the bottom of its swing.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

stored energy is weight times the height risen · the height risen by a pendulum comes from the geometry of the swing

VERDICT 72W

The ship is a body falling along a curved path. Mechanical energy depends on the change in height, not on the shape of that path. At 48 degrees, the centre of mass rises about 2.85 m above the bottom. Turning that height into kinetic energy gives about 7.5 m/s. The tower reaches the same square-root form by kinematics. The shared result is useful because both situations convert gravitational potential energy into motion.

TAKEAWAY 20W

A swing is a fall along a curve, and the height it falls through is the only part that matters.

END OF THE DAY 16W

Ranking work is a question about what each delay costs, not about what each fault is.

DAY 14 OF 15

What goes on the certificate

PLAN CARD 121W

Thursday, and the last day anything can be changed. Marcus Vey witnessed the drop test this afternoon and has said nothing since. 7 rides have a margin computed for them, 6 of those margins hold, and the coaster's is 1.20 against a certificate that has said 1.38 for 25 years. Marsh has 98000 pounds of advance bookings against the opening weekend. Today you set each ride against what has actually been derived for it, put a number on the ship's certified swing angle, and settle what the flume's boats may carry. A certificate is a claim in your name about what a machine will do to the next quarter of 1 million people through it, and it cannot be qualified afterwards.

BRIEFING 16W

The inspection is tomorrow, and a signature is not a thing that can be taken back.

WHAT YOU DECIDE 11W

Sign what the working supports, and say what it does not.

14.1 Seven rides and what can be claimed

THE DROP TOWER · DROP TOWER CONTROL · A RIDE · CASEBOOK

SCENE 36W

The safety engineer's file has one line for each ride. Four are ready to be filled in. The wording matters because the sentence on a certificate is the sentence an insurer or court will later read.

GUIDE 65W

Four rides on the left, and four forms of words on the right. Pair them by asking what the evidence behind each ride can carry. A tested claim covers the conditions actually tested. A margin that moves with performance needs a condition attached. A limit the machine enforces needs no person in it. And one of them has a known load and an unknown capacity.

ASK 10W

Match each ride to the claim its working actually supports.

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a claim is limited by the conditions the working was done under · a margin is a ratio between what is available and what is required

VERDICT 75W

Each certificate claim has to match the evidence that supports it. An observed tower test supports only the conditions actually tested. The coaster can be certified with a morning speed condition because its margin changes with performance. The carousel limit is different because the controller enforces it directly. The wheel has a known load but an unresolved defect capacity. That does not make the calculation useless. It identifies the exact missing piece that prevents certification.

TAKEAWAY 18W

A certificate is a set of conditions with a number attached, and the conditions are the harder half.

14.2 What the seat pushes with at the bottom

THE PIRATE SHIP · PIRATE SHIP CONSOLE · A PERSON · BALLPARK

SCENE 35W

At the bottom of the swing, the seat must support a rider and bend their path upward toward the pivot. The structural engineer wants the certified angle tied to seat force rather than to appearance.

GUIDE 64W

Five numbers, and one of them is the period, which does not enter this force. Ask of each whether the seat's job depends on it. At the bottom the seat carries the weight and bends the path upward as well, so the answer has two parts. And note how it grows: bottom speed rises with angle, so the seat force outruns the angle itself.

ASK 18W

Estimate the seat force at the bottom of a 48° swing, as a multiple of the rider's weight.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a body on a curve needs a net force toward the centre of that curve · at the bottom of a swing the seat pushes up and gravity pulls down

VERDICT 78W

At the bottom of the swing, the required centripetal acceleration points upward toward the pivot. Gravity pulls downward, so the seat must overcome weight and supply the extra inward force. Using v^2/r gives the added acceleration. The total seat force is about 1.67 times body weight at 48 degrees. This is what a rider feels as apparent weight. A larger swing angle produces a larger bottom speed, so the seat force rises faster than the angle itself suggests.

TAKEAWAY 16W

An angle is certified by the force it produces, not by the angle it looks like.

14.3 Four adults, and why not five

THE LOG FLUME · FLUME PUMPHOUSE · A RIDE · CHOICE

SCENE 34W

Fiona McCarthy still uses an inherited four-adult rule. The hulls changed two seasons ago. The certificate now has to state what a boat may carry, not simply what the old boats happened to carry.

GUIDE 62W

Four options, and one of them says nothing limits it at all. Ask of each whether floating and having enough side above the water are the same question. Five adults still float. What each one costs is height of hull above the waterline, and at splash-down the water rises to meet it. The old rule came from boats that no longer exist.

ASK 12W

What actually limits the number of adults in one of these boats?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a floating boat sits at whatever depth displaces its own weight · the drop at the end of the flume throws up a wave the boat meets

VERDICT 80W

Floating is not the same as having enough hull above the water. Five adults still displace enough water to float, so buoyancy alone does not set the operating limit. Each added passenger lowers the hull and leaves less side height above the water. At splash-down, the boat meets a wave and the water rises relative to the side. The useful limit therefore comes from that remaining height during the splash, not simply from whether the boat floats in calm water.

TAKEAWAY 16W

A floating body can still fail if too little of its side remains above the water.

14.4 Why the chains come out

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · CALLBACK · DERIVE

SCENE 35W

The swing carousel turns at ten revolutions a minute. Its chains are 4.5 metres long and hang from a hub 3.0 metres from the axis. An old chalk mark shows where the seats once reached.

GUIDE 58W

Two force equations start you off — one for the horizontal pull and one for the vertical. Build down a line at a time to a relation for the angle the chains hang at. Then read the relation itself: the useful part of this derivation is which quantities survive to the end and which cancel on the way.

ASK 19W

Work the two force equations down to a relation for the angle, and say what the angle depends on.

BEHIND THE BUTTON 166W

Why two equations. A seat on a chain travels a horizontal circle at constant speed, so the net force on it points at the axis and has no vertical part. Splitting the tension into its horizontal and vertical components gives one equation for each direction, and dividing one by the other eliminates the tension nobody measured.

What falls out, and why it matters at the gate. The mass cancels: a heavy rider and a light rider hang at the same angle, which is why the operator does not sort the queue by weight and why the old chalk mark is a mark about speed rather than about who was sitting there.

What the angle does depend on. The rotation rate and the geometry — chain length and hub radius, which together set the radius of the circle the seat travels in. So a carousel running slightly fast shows it in the angle, and the chalk mark is a record of what fast used to look like.

TAKES AS READ

a body moving in a circle needs a net force toward the centre · the tension in a chain acts along the chain

VERDICT 78W

The chain tension has two jobs. Its vertical component balances the rider's weight, while its horizontal component supplies the inward force for circular motion. Dividing those force equations removes both the unknown tension and the rider's mass. What remains links the angle to speed and the radius of the seat's circular path. Because the seat swings outward, that radius grows with the angle and must be solved with it. At ten rpm, the result is about 31 degrees.

TAKEAWAY 16W

Resolving one force into components can remove unknowns and reveal which variables actually control the motion.

END OF THE DAY 19W

A signature says what the working supports, and everything it does not cover has to be written down too.

What opens, and what is written down

BEFORE THE DAY — TAG

Catch the ride engineer before the tower test drop

The brake current settings from the last test drop are in one notebook, and once the tower is armed for a test nobody is walking up to it. This afternoon's argument about deceleration is against those settings, and the copy on the desk is last season's.

PLAN CARD 118W

Friday, and Vey has been on site since 13 of the 7 rides have a derivation, a number and a condition; the wheel has a computed bolt load and a weld indication that arrived too late to resolve. The park opens in eight days if the certificates are signed this afternoon. Today you decide what the winter's money goes on, state what can honestly be claimed about the coaster, and write the standing order for the bumper floor that Hart's staff will actually work to. Brennan came in at 9 with the tenth notebook, which he had found in a drawer, and it has the 1998 regrade in it — dated, with a sketch, and no drawing number.

BRIEFING 13W

The inspection is this morning, and what survives it is what was derived.

WHAT YOU DECIDE 11W

Say what has been established, how well, and what has not.

15.1 What to spend the winter on

THE COASTER · COASTER STATION · A RIDE · VALUE

SCENE 37W

There is money for only two winter jobs before next season. The park manager, Marta Kovač and the safety engineer each have a proposal. The Ferris wheel still has no final answer on its number-nine wheel joint.

GUIDE 63W

Two winter slots, three proposals. Open each one to see what its result could change: a certification, a design decision, or an argument nobody has to settle before reopening. Buy the two whose answers change what the park can decide, not the two that study the most interesting problem. The wheel joint has no final answer yet, which is a decision still open.

ASK 20W

You can fund two pieces of work. Which two buy information that changes what the park can decide before reopening?

BEHIND THE BUTTON 150W

What makes a piece of work worth funding. Not how much it would teach — how much it would change. Work that confirms something already certified produces a tidy report and no decision, and work that investigates a fault nobody can act on until next season can wait a season.

Why the wheel joint is the one to watch. An unanswered question about a joint that carries load is a question the park cannot reopen around. That makes information about it worth more than information about anything already signed off, whatever the relative engineering interest.

Why only two. A budget is what makes this a decision rather than a wish list, and the third proposal is not wrong — it is simply the one whose answer can arrive later without costing anything. Saying that out loud is the part that a manager has to be able to defend in spring.

TAKES AS READ

the wheel is the only ride still missing a certification decision · the coaster drawing has already been shown wrong at the top of the loop

VERDICT 82W

The wheel has a known joint load and a measured crack-like mark, but no crack-strength assessment. That missing result directly controls whether the wheel can be certified. The coaster has a different unknown because one important radius already disagreed with its old drawing. A full geometry survey could reveal other geometry errors. An overhaul may improve a known speed loss, but the morning rule already manages that condition. Repainting changes appearance only. With two work slots, the best purchases settle unresolved decisions.

TAKEAWAY 22W

> The best investigation is the one that can change a decision, not the one that merely adds detail to something already understood.

15.2 The honest sentence about number nine

THE FERRIS WHEEL · WHEEL MACHINE ROOM · A PERSON · CHOICE

SCENE 37W

Priya Raman has the computed bolt load. The inspector has the crack-like mark's measured length and depth. Nobody has the crack-strength assessment. The park manager needs one sentence for the board and insurers before the meeting ends.

GUIDE 67W

Four sentences for the board, and two of them claim a result nobody has produced. Ask of each which of the two facts it rests on: the load, or what the flawed joint can carry. The first is computed. The second needs an assessment nobody has done. Eighty-one seasons of service is not a margin, and the narrow version is what tells the specialist what to answer.

ASK 10W

What is the honest sentence about the wheel this afternoon?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a claim is limited by the working that supports it · an unresolved question is not the same as a negative finding

VERDICT 81W

The file already establishes the joint load and the size of the crack-like mark. It does not establish the crack's allowable load. Calling the wheel unsafe would claim a result the crack-strength assessment has not produced. Calling it safe would treat 81 years of service as proof of margin. Both overreach. The accurate statement keeps the two known facts and names the one unresolved question. That narrow uncertainty is useful because it tells the winter specialist exactly what must be answered.

TAKEAWAY 18W

> The strength of a claim is set by what was actually established, and saying so is the claim.

15.3 A rule for the four-hundredth cycle

THE BUMPER CARS · BUMPER CAR PAVILION · A RIDE · SEQUENCE

SCENE 35W

The bumper-floor supervisor is writing a standing order for operators who will not have read this fortnight's calculations. Four recurring conditions are on the sheet. Each needs a response that can survive a busy shift.

GUIDE 68W

Four instructions, and each answers a different failure. Ask of each what would go wrong if it came later. A floor speed set too high raises the energy of every collision on the cycle, so it belongs before the gate. An unlatched belt is one seat. A car that stops taking power is one car. And a record is what stops the next shift returning it to service.

ASK 11W

Order the four instructions for one cycle on the bumper floor.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

kinetic energy grows with the square of speed · a fault on one bumper car can be isolated without closing the whole floor · velocity against speed: direction is part of the quantity — taken as read

VERDICT 72W

Each response belongs to a different failure. Excess floor speed raises collision energy because kinetic energy scales with v^2, so the common setting must be corrected before dispatch. An unsecured restraint is a rider-level problem and that seat cannot run. An intermittent electrical-contact fault belongs to one car, which can be isolated. A removed car also needs a persistent record. Otherwise the next shift can unknowingly return a known fault to service.

TAKEAWAY 22W

A standing order should turn a physical limit into an action that a new operator can carry out without reconstructing the calculation.

15.4 What a pole holds sideways

THE CAROUSEL AND SWINGS · CAROUSEL DRIVE HOUSE · A RIDE · CALLBACK · BALLPARK

SCENE 34W

An outer horse carries a 78 kilogram rider 6.2 metres from the axis. The carousel is turning at eight revolutions a minute. Bisi Adeyemi wants the sideways force before anyone raises the controller setting.

GUIDE 65W

Five numbers, and two of them belong to earlier or other steps: the turns a minute, and gravity. Ask of each whether this force depends on it. Turns a minute is not a speed, so one number here is already the conversion done for you. And note the square: nine turns instead of eight is twelve per cent more speed and a quarter more force.

ASK 18W

Estimate the sideways force on the rider, converting the carousel's eight revolutions a minute into linear speed first.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a body in a circle needs a net force toward the centre · that force is the mass times the centripetal acceleration

VERDICT 76W

Eight revolutions a minute is an angular rate, not the rider's linear speed. One revolution covers 2πr, so the rider moves about 5.2 m/s at 6.2 m radius. Then $F = mv^2/r$ gives about 340 N toward the centre. The square on speed makes controller changes expensive. Going from eight to nine rpm raises speed by 12.5%, but the required inward force by about 27%. The pole and saddle must supply that extra force every turn.

TAKEAWAY 20W

Circular-force problems often hide the needed speed inside a rotation rate, so convert the motion before applying $F = mv^2/r$.

END OF THE DAY 17W

What outlives this fortnight is what somebody wrote down in a form the next person can use.

THE CLOSING CARD

Six of the seven opened. The tower went first, on a witnessed drop that logged 5.4 g against the 6.0 the certificate allows; the carousel and the swings opened on Brennan's own settings, which the derivations confirmed to within a degree; the ship opened after its drive was retimed from 5.60 seconds to the 5.90 the pendulum actually keeps, which Sam Idowu had been describing from the floor for two seasons. The coaster opened with a condition on it — a minimum station return speed, read with a wheel every morning before the first train — because its loop was regraded in 1998 and the crown demands 8.5 metres a second rather than the 7.4 the 1974 drawing implies.

What it cost: the wheel stayed shut. The load in arm nine's bolt group is computed at 52.7 kilonewtons and whether a 41-millimetre indication can carry it is a fracture assessment nobody on site could do in three weeks, so twenty-four gondolas stood still through the whole season and Marsh borrowed against next year to cover it. What is unfinished: the coaster's margin is 1.20 and falling about a tenth of a metre a second each season, so the morning rule buys time rather than settling anything; the flume's pump runs at 53.7 kilowatts against a 55-kilowatt plate; and the 1974 drawings are still the only ones the park has, now known to be wrong about at least one dimension by 1.8 metres.

Card text only — no options, boards or answers. 12,780 words of card prose across 61 stops. Generated from `themes/midway` by `tools/make-cardbook.mjs`.